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Outcome Tests.*

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Abstract

Outcome tests promise to detect discrimination without observing all the information on which decisions are based. In the setting of police stops and searches, their traditional Beckerian motivation compares outcomes among marginal motorists: those for whom the officer's belief about finding contraband makes her indifferent. Yet conventional approaches require strong assumptions to infer discrimination from outcomes, while being unable to quantify how much discrimination there is. I argue that targeting these elusive marginal motorists is unnecessary and limits the scope of the test. To examine these limitations, I develop the first fully nonparametric causal framework for the outcome test, providing a unifying basis upon which to evaluate previous statistical and formal approaches to the test. I will introduce novel graphical representations of officer' beliefs and taste-based and statistical discrimination. The framework will reveal how collider bias explains why conventional approaches perennially suffer from making strong assumptions on behavior and rationality, clarifying how inferences from prior work depend on the accuracy of officers' beliefs. To move beyond these limitations, I reimagine outcomes as negative controls for discriminatory decisions, opening a new style of thought for testing and quantifying discrimination. Drawing on ideas from proximal causal inference, I derive sharp bounds on the probability of necessity—the fraction of minority searches that would not have occurred but-for their race—endowing the outcome test with a measure of disparate treatment. I illustrate the approach with an application to searches by the Washington State Patrol.

Keywords: outcome test, partial identification, proximal inference, mediation, statistical discrimination, collider bias

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1 Introduction

Discrimination is a causal phenomenon. A decision is discriminatory when group membership would change how an individual is treated, holding fixed the features of the situation that merit the decision. This description is inherently counterfactual, and thus understanding discrimination as the mechanism for producing observed disparities in high-stakes settings such as police enforcement, bail decisions, hiring, lending, cannot be separated from a causal treatment of the empirical problem.

However, discrimination rarely manifests in a directly observable way. Decision-makers act in settings where the relevant facts, their beliefs about those facts, and the reasoning connecting beliefs to action are only partly and biasedly observed by both themselves and the analyst. For example, in traffic enforcement, officers biasedly respond to features of the driver, the vehicle, the roadway, the neighborhood, among others, only some of which appear in administrative data. In hiring, admissions, or pretrial release, decision-makers act on applications, records, scores, interviews, and subjective assessments whose substantive content is itself an unreliable measure of merit for the relevant judge's decision.

Two broad families of observational tests for discrimination have been designed to confront this problem: benchmark tests and outcome tests. Benchmark tests attack the problem directly, by attempting to construct a comparison population of similarly situated individuals in which discrimination is absent. In its simplest form, the analyst compares decisions across racial groups after adjusting for observed merit; in traffic enforcement, more elaborate designs compare the racial composition of police stops to an external population intended to represent motorists legitimately at risk of being stopped, such as licensed drivers, crash-involved motorists, roadside observations, or traffic-camera records (Rojek et al., 2004; Engel et al., 2002; Alpert et al., 2004; Lange et al., 2005). However, if the characteristic that legitimately affects the decision is unobserved, is associated with race, and is not adequately represented by the chosen benchmark, the resulting racial comparison is subject to omitted-variable bias, where the majority of studies in this area fall down (Cooper et al., 2026b).

Outcome tests were postulated in an effort to sidestep this problem. Rather than attempting to observe every circumstance that legitimately enters the decision, they use the outcome of the decision to learn about them indirectly. The idea originates with Becker's theory of discrimination (Becker, 1971, 1993). Consider a police officer deciding whether to search a vehicle for contraband. Suppose the officer searches when her subjective likelihood of discovering contraband exceeds some threshold. A lower threshold for minority motorists constitutes a lower evidentiary standard for search: a less suspicious minority driver is more liekly to be searched than an otherwise comparable White driver. Becker's insight was that this difference should reveal itself in the productivity of decisions at the "margin" (of indifference). If there exist motorists for whom the officer is just indifferent between searching and not searching—the *marginal motorists*—then, absent differential standards, the likelihood of finding contraband among these motorists should be equal across racial groups. This reasoning shows promise for revealing discrimination without requiring the analyst to directly observe all of the information upon which the decision was based.

Turning this intuition into an empirical test has occupied a substantial literature, primarily in economics. There, it is emphasised that the chief practical problem with Becker's intuition is that the marginal motorists are not observed. Analysts instead observe average *hit rates*: the rate at which contraband is discovered among all motorists who were searched. Differences in these averages may fail to reflect differences at the margin of the decision because the distribution of motorists away from the margin may differ across race, the well-known problem of inframarginality (e.g. Ayres, 2002). A sequence of influential contributions has sought to overcome this difficulty. These include game-theoretic equilibrium models impose additional behavioral structure under which average hit rates become informative of bias (e.g. Knowles et al., 2001; Anwar and Fang, 2006); the invocation of invoke strong restrictions on the heterogeneity of background qualifications across groups (Gaebler and Goel, 2025); the specification of a highly parametrized model for the decision-making process (Simoiu et al., 2017; Pierson et al., 2020), and other approaches exploit variation

across decision-makers to recover outcomes at the decision margin directly, avoiding inframarginality altogether, by appealing to a marginal treatment effect framework (Heckman and Vytlacil, 2007) through the use of an instrumental variable (Arnold et al., 2018, 2022; Hull, 2021; Canay et al., 2024).

Concurrently, since Becker formulated the outcome test, other economists postulated that Becker's proposed mechanism of discrimination, which has been termed a 'taste' for discrimination, may not be the only explanation for observed disparities. Seminal work by Phelps (1972) and Aigner and Cain (1977) proposed a theory of statistical discrimination, the idea that officers receive an incomplete signal regarding the presence of contraband in the vehicle, and use their priors about how race correlates with carrying contraband to improve upon this signal, for instance by stereotyping (Arrow, 1973; Bordalo et al., 2016). Widely accepted in economics as a valid mechanism producing discrimination, the marginal outcome test as pitched by Becker cannot distinguish the two mechanisms, as the simple formulation of the test renders them observationally equivalent. Economists, then, have additionally striven to develop methods which render the hit rates informative for separating taste from statistical discrimination, by first eliminating the problem of inframarginality.

Now, almost all studies implementing the outcome test motivate their methodology using Becker's original insight, yet, there is surprisingly little unity in how the outcome test is implemented. Different approaches formalize the decision problem in substantially different ways, and concepts that appear common at an intuitive level do not materialize as common restrictions on the data-generating process. Identification strategies vary considerably: some models eliminate heterogeneity among the individuals facing the decision; others restrict heterogeneity among decision-makers. Most, albeit implicitly, rely on decision-makers' beliefs to be calibrated to observed outcomes.[1] Still others recover tractability through heavily parameterized models of decision-making. Invoked assumptions are often taken as primitives, acknowledged only indirectly, or left implicit. At the same time, almost all studies do not

[1] A belief is *calibrated* when it coincides with the true conditional probability of the outcome given the information on which it is based. I give a precise statement of this notion in Section 4.2 (Definition 4).

begin by specifying a causal estimand that quantifies discrimination. The inferential target is instead framed in terms of whether a unobserved decision standard differs across groups, leaving unclear how the resulting conclusion maps onto the prevalence or magnitude of discriminatory treatment.

I argue that this fragmentation stems from the manner in which the outcome test has been conceived. The outcome test has traditionally been motivated as a method for detecting differences in the standards applied to marginal individuals. Once this is taken as the target of the exercise, the analyst must somehow recover or make inferences about an inherently unobservable object, whose existence is taken for granted. Inframarginality bias arises as a consequence of this difficulty, but is not itself the core of the issue. More fundamentally, the Beckerian framing of the test both obscures the assumptions needed to connect observed outcomes to discrimination and discourages the specification of a causal estimand that directly describes the discriminatory behavior of interest.

The thesis of this article argues for a different interpretation of the outcome test, abandoning marginal standards as the target of inference. In the context of traffic stops and searches, the analyst may not observe all circumstances that made a vehicle worth searching, such as the odor of marijuana, but those same circumstances should also make contraband more likely to be discovered. Successful searches thus leave a noisy trace of the legitimate reasons for searching the vehicle. If race causes officers to search motorists who would not otherwise have been searched, those motorists should, under appropriate restrictions, carry weaker traces of those legitimate reasons. This follows because of an exclusion restriction: race cannot directly influence the discovery of contraband in the car, except through its influence on the decision to search. This reasoning reimagines the eponymous outcome of the test as an outcome proxy, or negative control outcome (Lipsitch et al., 2010), for the features of the stop that induce a search. In so doing, by appealing to ideas from proximal causal inference (Tchetgen Tchetgen et al., 2024), I introduce a new class of identification strategies that utilise the intuition of the outcome test for the detection of discrimination.

I develop this argument by introducing the first fully nonparametric causal model describing the outcome test, whence this new interpretation of the test arises. In this article, I will ground the exposition in the empirical setting of police searches for contraband as it is sufficiently general to capture much of the nuances of the test. Section 2 specifies this model, showing how it implies general causal graphical notion of the officer's belief about the presence of contraband, without loss of generality. An immediate consequence of this result is that the threshold model assumed primitive may reassuringly be imposed without loss. Equipped with this notion of belief, I introduce novel, highly general definitions of taste-based and statistical discrimination based on the theory of causal mediation, and discuss issues of identification.

Before revealing how this framework naturally gives rise to a new interpretation of outcome tests, I first demonstrate its generality by re-examining what can be learned from conventional hit rate comparisons, through the lens of my proposed framework. Section 3 confirms that racial differences in average hit rates are generally uninformative about differential standards without restrictions on unit-level interactions induced by selection into the decision. I use familiar causal ideas like principal stratification, conditioning on a collider, and causal interaction, to intuitively pinpoint both the source of this problem and the assumptions under which prominent hit rate tests become informative. I then revisit several influential theoretical models of the outcome test, showing how each controls sources of ambiguity through some combination of game-theoretic reasoning, homogeneity restrictions on motorists or decision-makers, calibrated beliefs, and parametric assumptions. In Section 4 I subsequently turn to so-called marginal outcome tests and argue, although they address inframarginality directly, connecting observed outcomes at the margin of the decision to the decision-maker's standard requires strong assumptions on their beliefs, and is identified in seldom available conditions.

Next, in Section 5, I reveal the alternative interpretation of the outcome test suggested by the causal model, diverting attention away from differential standards. I then show how

this sort of reasoning engenders a new class of identification strategies for a chosen causal estimand. This article situates one such choice: the probability of necessity, the probability that a searched minority motorist would not have been searched had that same motorist been perceived as White. This target directly quantifies the prevalence of discriminatory searches and adorns the outcome test with a causal interpretation that is meaningful for policy and closely connected to legal notions of but-for causation (Dawid et al., 2014, 2017; Cuellar, 2022). In Section 6, I apply the resulting method to police stops and searches in Washington state. I further examine heterogeneity in this quantity across geography, and the policy change which legalized marijuana throughout the state, illustrating how an explicitly defined causal estimand can generate substantively sharper conclusions than conventional comparisons of search or hit rates.

I remark that I am not the first researcher to provide a formal basis for understanding when outcome tests are informative; recent work by Canay et al. (2024) and Hull (2021) outline some important theoretical foundations. However, this article goes beyond their analysis by generalising the decision-making model and re-examining the very foundation of what motivates the test. Section 7.1 reveals how the model defined here contains their descriptions of the decision-making process and discuss further connections between our ideas. I then describe the potential for future work based on the model I propose in this work in Section 7.2. The article subsequently concludes.

2 Causal Model

I present a model for the data-generating process (DGP) underlying the officer's decision to search a vehicle following a traffic stop using the language of nonparametric structural equation models (Pearl, 2009). In Section 7.1, I will be precise about the advantages of this approach for specifying the problem setting relative to common economic thinking; here, I start with technical details. The running empirical setting in which I build the model is

searches for contraband during traffic enforcement, as it captures most of the nuances that create the challenges underlying the test. Throughout, the 'decision-maker' will be a police officer, however, I will use this term, interchangeably with 'judge', more generally to discuss decision models in other settings.

Let Ω be a set of traffic stops in a given jurisdiction; the unit of analysis $\omega \in \Omega$ is a traffic stop. Define binary variables, $R(\omega) \in \{r_0, r_1\}, D(\omega) \in \{d_0, d_1\}$ and $Y(\omega) \in \{0, 1\}$, the minority status of the driver, the decision to search and the discovery of contraband respectively. As $Y(\omega)$ denotes the discovery of contraband, $Y(\omega) = 0$ whenever $D(\omega) = d_0$. Let $X(\omega)$ be non-race characteristics of the encounter that are observed to the analyst, such as location and time of the stop, vehicle condition, reason for stop etc. Denote by $U(\omega)$ characteristics which are unobserved by the analyst and potentially observed by the officer, such as the officer's mood. I collect these contextual elements into a vector $V(\omega) = (X(\omega), U(\omega))$. Notably, my analysis will assume nothing about the structure of $V(\omega)$. Used only to emphasise the unit of analysis, the argument ω will henceforth be suppressed unless that particular emphasis again serves the exposition.

The variables above are governed by the following causal model

$$R \leftarrow f_R(U_R), \quad V \leftarrow f_V(U_V), \quad D \leftarrow f_D(R, V, U_D), \quad Y \leftarrow f_Y(D, V, U_Y), \tag{1}$$

where U_W represents an exogenous disturbance associated with variable W. In words, during a traffic stop, an officer decides whether to conduct a search for contraband based on features of the motorist and the environment, some of which is partially observed by both the officer and the analyst. Upon deciding to search the vehicle, contraband may be found. Race is assumed not to affect contraband discovery except through the decision to search. Thus, holding fixed whether a search occurs and all other features of the encounter, changing the motorist's perceived race cannot change whether contraband is discovered. This exclusion restriction does not preclude observed racial differences in discovery rates, which may arise

through backdoor confounding by V.[2]

Often, debates surrounding the necessary inclusions of certain variables in the decision-making model centers on what information the analyst can credibly model the decision-maker as using to make the decision (see for instance, Bohren et al. (2025), or the debate between Canay et al. (2024) and Hull (2021)). This delineation matters for understanding the mechanisms by which discrimination arises. In model (1), I am deliberately general in that V contains all environmental features about the stop that could influence a decision to search, whether or not they are observable to the officer. This is important to specify because while it may not be so that V wholly influences D, its effect on Y must be accounted for in any empirical analysis. Contextual features of the stop that the officer does not perceive cannot influence the decision to search. The features the officer does perceive matter only insofar as they inform the officer's subjective belief that a search will uncover contraband, and it is this belief that forms the basis of the decision. Next, I show how this degree of belief over whether the officer will find contraband may be added to the causal model, without loss of generality.

I augment the causal model in the following manner

$$\begin{aligned} R &\leftarrow f_R(U_R) & V &\leftarrow f_V(U_V) & B &\leftarrow f_B(R, V, U_B) \\ D &\leftarrow f_D(R, B, U_D) & Y &\leftarrow f_Y(D, V, U_Y) \end{aligned} \tag{2}$$

where we will interpret the variable B as officer's 'belief' about the discovery of contraband in the vehicle. Remark that in this model we now assume that V does not directly cause D. That is, an officer's decision to search the vehicle depends on the environment V insofar as it influences their belief about contraband in the vehicle. This restriction is often formulated in the economic literature as officers receiving a 'signal' about the outcome Y, upon only

[2] A conceivable violation would occur if officers conducted more intensive or thorough searches of minority motorists, so that perceived race affected whether existing contraband was detected even conditional on a search. I treat this as a comparatively limited concern: once an officer elects to search a vehicle, professional competence should make discovery depend principally on whether contraband is present, leaving less scope for marginal differences in search intensity to determine the outcome.

which they act in the absence of discrimination. This exclusion could be violated if an officer has been informed that a specific aspect of V should induce a search prior to pulling over the vehicle, limiting their ability to exercise discretion for searching such vehicles. For example, suppose in a particular day's assignments officers had been told to concentrate on red hatchbacks. If minorities are more likely to drive red hatchbacks, then violation of this exclusion would imply that discrimination is indistinguishable from officers' non-discretionary use of their private information. In this example, failing to control for the prevalence of red hatchbacks ownership by race would mean that the test a-priori could not logically identify discrimination. Examining the extent to which this behavior is prevalent within any empirical study requires close engagement with the obtained data and use of domain expertise. Going forward, I maintain this exclusion restriction but discuss it further in Section 7.1 by situating this point alongside a similar observation made by Canay et al. (2024).

Currently, I have made no assumptions on the nature of B nor its state space. Were B in fact high-dimensional, it would not be amenable to an interpretation as a belief or internal prediction. Next, I will show that we can—under a mild regularity condition—define B as the officer's subjective likelihood for finding contraband in the vehicle, a scalar on the unit interval.

Proposition 1 (A graphical notion of belief) *Consider the causal model* (2) *with the state space $\mathcal{B}$ of B unrestricted.*

(i) Canonical beliefs. *Fix r. Define an equivalence relation $\sim_r$ on $\mathcal{B}$ by $b \sim_r b'$ if $f_D(r, b, U_D) = f_D(r, b', U_D)$ almost surely. Let $B^\dagger(r) := [B(r)]_{\sim_r}$ denote the resulting equivalence class, named the canonical belief. Then there exists a function $f_D^\dagger$ such that*

$$D(r, B(r), U_D) = f_D^\dagger(r, B^\dagger(r), U_D) \qquad a.s. \tag{3}$$

(ii) Subjective probability. *For each canonical belief $b^\dagger$, define $A_r(b^\dagger) := \{u_d : f_D^\dagger(r, b^\dagger, u_d) =$*

$d_1\}$. *Assume that these sets are nested up to sets of* U_D*-measure zero: for every* $b^\dagger, b^{\dagger\prime}$*, either* $A_r(b^\dagger) \subseteq A_r(b^{\dagger\prime})$ *almost surely or* $A_r(b^{\dagger\prime}) \subseteq A_r(b^\dagger)$ *almost surely. Then there exists an injective scalar representation* $\phi_r : \text{supp}(B^\dagger(r)) \to [0,1]$ *that preserves this ordering. Writing* $\widetilde{B}(r) := \phi_r(B^\dagger(r))$*, there exists a threshold* $t(r, U_D)$ *such that*

$$D(r, B(r), U_D) = \mathbf{1}\{\widetilde{B}(r) > t(r, U_D)\} \qquad a.s., \tag{4}$$

up to the convention governing the decision when $\widetilde{B}(r) = t(r, U_D)$*.*

Proof. See Appendix A.1. ■

The above result is mildly technical however it can be described simply. In part (i), since the decision to search is binary, whatever the state space of B, conditional on race r and officer type u_d it may be organised into regions that induce a search, or those that do not. This argument is similar in nature to the canonical partitioning procedure underlying the literature on partial identification using principal stratification (Balke and Pearl, 1997; Duarte et al., 2024). For the purposes of producing a search of the vehicle, separating beliefs into these regions of search and no-search is without loss for the data-generating process.

Part (ii) is what permits me to assign meaning to the reorganised beliefs inside those partitions. This meaning is derived by constructing a coherent manner in which to order the beliefs. To accomplish this, I leverage a minor regularity condition which is the nesting of the sets of officer types that lead to a search for a given belief, $A_r(b^\dagger)$. This assumption is necessary for the following reason: suppose that officer i searches at belief $b_0^\dagger$ and not at $b_1^\dagger$, while officer j does the reverse. Such a scenario would make it impossible to meaningfully order $b_0^\dagger$ and $b_1^\dagger$ in their ability to induce a search, prohibiting the existence of any scalar representation of $B^\dagger$.[3] The nesting condition forbids this behavior. Crucially, this condition does not force officers to rank motorists identically, rather, it requires that officers respond similarly of the strength of their own beliefs.

[3]In economics, this kind of restriction on behavior is called a no-crossing condition (Vytlacil, 2002).

To see this concretely, consider the following analogy: this year in PSCI 101, Professor i grades essays of Alice and Bob at 60% and 72% respectively. The class grading is moderated by Professor j to ensure consistency, but she respectively grades these students at 80% and 45%. Analyzing their grading data along with their own preferences, to pass in Professor i's eyes, students require a score exceeding 70%, but for Professor j students need to achieve 65%. Applied here, the nesting condition requires that Professor i passes Bob and Professor j passes Alice. Note, the Professors do not agree on whose essay was better; the assumption I use above requires that there be a coherence between Professors on the meaning of the score as it pertains to passing the assignment. In causal terms, heterogeneity in U_B remains unrestricted, so different officers may order the perceived the guilt of different motorists as they like. However the nesting condition requires that conditional on these beliefs, U_D may alter how willing an officer is to search according to their beliefs, but may not reverse that ordering. This is a weak behavioral requirement on the data-generating process.

Finally, because $\widetilde{B}$ lies on $[0, 1]$, and $Y(d_1)$ is binary, we are free to interpret it as the officer's subjective degree of belief that contraband would be found in the vehicle if searched, almost entirely derived from the causal model alone. This is in contrast to past work, where such threshold models have been assumed as primitive with additional modeling restrictions, for example, where it was taken for granted that $\tilde{B}$ may exist without conceptual loss (Canay et al., 2024; Hull, 2021; Simoiu et al., 2017). As the construction in the proposition above shows, the scalar prediction $\tilde{B}$ completely mediates the path from B to D, so going forward I relabel $\widetilde{B}$ to B for legibility.

2.1 The Dichotomy of Discrimination

A central difficulty in the empirical literature on discrimination is that the quantity of substantive interest is not always the quantity directly estimated. The motivating question is typically one of disparate treatment: whether, or how often, a decision would have differed for the same individual had their race been different. Because this question is inherently coun-

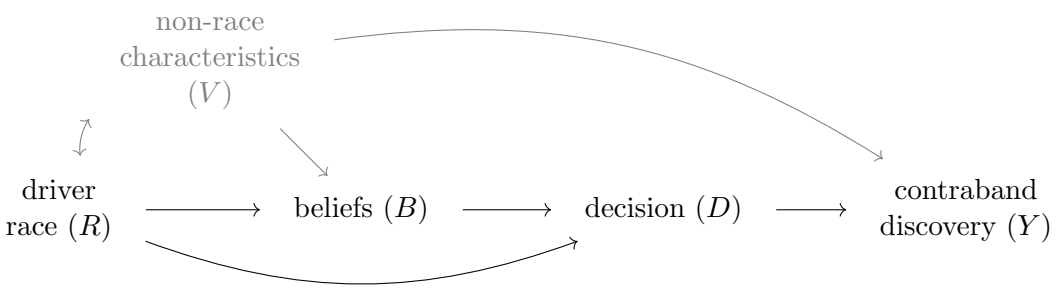


Figure 1: **Causal graph for contraband search.** Non-race characteristics about the encounter inform an officer's beliefs about the likelihood of discovering contraband, possibly leading to a search. Disparities in the effect of R on D decompose into taste-based discrimination ($R \to D$) and statistical discrimination ($R \to B \to D$). Exogenous disturbances are suppressed.

terfactual, it is difficult to answer using administrative records. Researchers have therefore often settled for quantities describing the disparate impact of decisions or other observable racial disparities, such as differences in decision rates under some equalizing condition (e.g. Knowles et al., 2001; Simoiu et al., 2017; Arnold et al., 2022). Such disparities may provide evidence about disparate treatment under additional assumptions, but frequently fall short of doing so. This has produced inconsistent framings of discriminatory behavior across the literature: an observed disparity is variously described as discrimination itself, as a test for discrimination, or as a consequence to be explained by taste-based or statistical discrimination. While not pernicious in and of itself, it obscures which identification assumptions license each conclusion, because the models used to represent decision-making differ across studies in consequential but often non-obvious ways. I instead begin by defining disparate treatment as a causal property of the decision (e.g. Knox et al., 2020) and subsequently distinguish the mechanisms through which it may arise.

Definition 1 (Racial bias) *An officer during encounter ω is said to exhibit racial bias against minorities if $\psi(\omega) \coloneqq D(r_1)(\omega) - D(r_0)(\omega) > 0$. An officer is said to exhibit anti-White bias in encounter ω if $\psi(\omega) < 0$.*

Many estimands may be used to quantify the extent of discrimination; however, each pertains to this consistent graphical notion. I note here that this formalization of racial bias is not in

conflict with traditional approaches in economic modeling. Economic models may differ in the mechanisms through which discrimination arises and may impose additional restrictions that permit those mechanisms to be separately identified. But these are refinements of the underlying causal claim, which is succinctly represented by the graphical model (2) through Definition 1.

Since the seminal work of Becker (1971), economists have striven to delineate two types of behavioral phenomenon that produce observed disparities by race. These are: taste-based and statistical discrimination. The former kind has been described as underlying Becker's original model of discrimination in the labor market, where employers would incur a cost or disutility from hiring a member of an underrepresented or marginalized group. This cost was modeled as completely separate from other qualifying characteristics for the decision, which is why this type of discrimination has also been described as an 'animus' against the marginalized group. Alternative theories have arisen where discrimination is modeled as a rational consequence of decision-makers solving an optimization problem under imperfect information. Employers seeking to maximise future productivity with only knowledge of an individuals interview performance and CV, for instance, are theorised to use correlates of important unobserved information, such as race or gender, to improve their prediction of future work performance. Should distributional differences exist between individuals by group conditional on unobservable qualifications, economists have argued that discriminating by group membership is a rational choice for maximising productivity. Due to the nature of this inference, this phenomenon is known as statistical discrimination. For in-depth reviews of these concepts as they are used in economics, political science and sociology see (Lang and Lehmann, 2012; Guryan and Charles, 2013; Tilcsik, 2021).

Distinguishing these types of discrimination has largely been an academic pursuit. As both phenomena are illegal discriminatory practices (Supreme Court of the United States, 2017)—as race is directly used by the judge to influence the decision—it is often of primary public interest to detect evidence for and the extent to which racial bias occurs, as indicated

by Definition 1, along with moderating factors. However, it has been argued that understanding the prevalence of statistical discrimination is important for designing interventions which edit decision-makers (likely erroneous) beliefs about a marginalized group (Bohren et al., 2025), so that one might in future reduce discrimination. Next, I will discuss how these phenomena can be described within my proposed causal model within the context of traffic stops, where they can be faithfully represented as path-specific effects (Avin et al., 2005; Malinsky et al., 2019; Pearl, 2001; Robins and Greenland, 1992; Pearl, 2001; Imai et al., 2011) through and avoiding belief, B.

Definition 2 (Taste-based discrimination) *An officer during stop ω is said to exhibit taste-based discrimination against minorities during stop ω if*

$$\psi_t(\omega) \coloneqq D(r_1, B(r_1))(\omega) - D(r_0, B(r_1))(\omega) > 0. \tag{5}$$

Definition 3 (Statistical discrimination) *An officer during stop ω is said to be exhibiting statistical discrimination against minorities whenever*

$$\psi_s(\omega) \coloneqq D(r_0, B(r_1))(\omega) - D(r_0, B(r_0))(\omega) > 0, \tag{6}$$

First, consider Definition 2. In words, for a given stop ω of a minority motorist, where the officer forms beliefs about the presence of contraband on this perception, ψ_t measures if their decision would change were the motorist perceived as White. Why does this definition capture taste-based discrimination? I offer two reasons. By fixing $B(r_1)$, ψ_t explicitly removes the effect of race through beliefs, isolating the natural direct effect (NDE) of race on the decision. Specifically, this aligns with the idea that taste-based discrimination reflects differential treatment due to race alone, aside from the prediction problem the decision-maker aims to solve. Second, in the threshold representation (4), it follows immediately that $\psi_t = 0$ if $t(R)$ is a constant function of R, which is consistent with nearly all previous economic models' definition of taste-based discrimination.

Next, consider Definition 3. In words, $\psi_s(\omega) > 0$ whenever, for a given stop ω of a White motorist, if the officer's belief about the likelihood of discovering contraband were changed from that associated with a White driver to that associated with a Black driver, holding the motorist fixed, their decision would change. Complementarily to taste-based discrimination, this definition describes statistical discrimination as the natural indirect effect (NIE) of race on the decision. This choice directly encodes the canonical description of statistical discrimination: the use of group membership as a proxy for unobserved characteristics. In this framework, this occurs precisely through the path $V \leftrightarrow R \rightarrow B$, whereby race shifts beliefs about V. The contrast ψ_s isolates this channel by holding race in the decision rule fixed while varying only the belief induced by race. Importantly, ψ_s is agnostic to the calibration of the officer's belief: it is neither implicit nor assumed. Therefore, should B be modeled as calibrated, ψ_s is so-called accurate statistical discrimination and inaccurate otherwise (Hull, 2021; Bohren et al., 2025).

Definitions 2 and 3 are novel and unify the classical Beckerian and Phelpsian theories of discrimination. Their advantages are fivefold. First, expanding the notion of racial bias Definition 1, for an encounter ω, $\psi(\omega) = D(r)(\omega) - D(r')(\omega) = \psi_t(\omega) + \psi_s(\omega)$, providing a clear decomposition of the mechanisms that could engender racial bias. Second, ψ_t and ψ_s are sufficiently general that their representation within some specific model for an analyst's specific task is straightforwardly done by imposing additional structure on B, a quantity which Proposition 1 shows exists without loss. Third, as those familiar with mediation analyses in causal inference are well aware, these definitions also reveal the difficulty in credibly delineating these effects in studies with human decision-makers. Indeed, B is not measured, and considering human decision-makers is at best approximated with proxies (likely subject to measurement error), adding additional complexity atop the fact that the NIE and NDE are generally not identified in causal models without strong assumptions (Imai et al., 2010).

Fourth, these definitions mark a departure from much of the existing literature on mecha-

nisms of discrimination which theorizes statistical and taste-based discrimination as competing explanations for an observed disparity. Indeed, the initial debate the progenitors of the theory of statistical discrimination, posed their model in conflict with that of Becker (Phelps, 1972; Aigner and Cain, 1977). Seminal work in this domain has encouraged an approach for testing for the presence of statistical discrimination as what must remain when taste-based discrimination is absent (e.g. Knowles et al., 2001; Anwar and Fang, 2006), if there is any remaining observed disparity by some measure—in the context of stops and searches, this is typically a disparity in search rates by race. However, by not identifying a precise notion of statistical discrimination as Definition 3 would offer, this retrospective line of reasoning may struggle to distinguish statistical discrimination from other causal phenomena such as standard confounding. I will demonstrate how this inferential error may arise in Section 3.

In fact, the literature on outcome tests has largely ignored the notion that taste-based and statistical discrimination could be operating simultaneously, and the effect of this on the magnitude of racial bias. For example, in Hull's formalization of the outcome test, he defines accurate statistical discrimination in such a manner that if it is operating then mechanically taste-based discrimination cannot occur (Hull, 2021). To give some intuition for why I reject the notion that these mechanisms must compete, consider the following two officer-civilian encounters during a traffic stop. The first has a civilian with characteristics $v_0 = (\text{Black}, \text{drives a BMW}, \text{polite}, \text{wears a suit}, \text{driving on Park Avenue})$, while the second is nearly identical but has $v_1 = (\text{Black}, \text{drives a BMW}, \text{polite}, \text{wears a tracksuit}, \text{driving in Harlem})$. Suppose the officer has a fixed preference for searching Black drivers, regardless of the other features of the encounter, so that ψ_t does not vary across v_i. However, one might expect $\mathbb{E}[\psi_s \mid V = v]$ to be much smaller under $v = v_0$ than under $v = v_1$, because the surrounding features of the encounter may alter the salience of racial stereotypes. Due to the reputation of the night-time activities of certain industry professionals who live on Park Avenue (United States Senate, 1988, p. 11874;The Wall Street Journal, 1983; Abelson and Smialek, 2017; Gordon, 2022), a Black man in a suit driving a BMW on Park Avenue may induce relatively

little additional suspicion compared with an otherwise comparable White man, whereas the same racial contrast may generate a much larger difference in beliefs when the driver is wearing a tracksuit in Harlem. In this example, the officer may therefore be less likely to conduct a discriminatory search under v_0 than under v_1, because the undeniably active statistical discrimination reinforces the taste-based component much less strongly. The calibration of officer beliefs in either scenario does not invalidate this observation. By defining these discriminatory mechanisms prospectively, may in theory study to what degree the strength of each of these components of racial bias affects the magnitude of measured disparities.

Lastly, the primitive level at which taste-based and statistical discrimination are defined forces the analyst to be clear about the calibration of beliefs B in their empirical strategy. Recent work has shown that what is known as inaccurate statistical discrimination can be observationally equivalent to taste-based discrimination (Bohren et al., 2025). Such a fact is made obvious by Definitions 2 and 3, because B (and parts of V) are unobserved, meaning that inferences based on statistical quantities which do not target B directly are subject to this problem. Precision about which phenomena are generating the observed data—taste-based, accurate statistical, or inaccurate statistical discrimination or confounding—would require strict restrictions on the data-generating process. Adhering to the above causal model encourages transparency on this point for any empirical analysis.

This subsection founds my causal framework for the outcome test. In Sections 3 and 4 I reveal how the framework can be used to simply understand why the outcome test is exceptionally delicate absent often unjustifiably strong assumptions.

3 Hit Rates Condition on A Collider

The outcome test generally comes in two varieties, both derived from the Beckerian intuition for the test. Becker posited that among the searched units for whom the officer was indifferent, their risk of contraband discovery must equal the officer's indifference threshold in the

absence of discrimination. Should this not occur, it might imply that officers are applying a double standard, which is typically specified as $t(r_0) \neq t(r_1)$ as in (4). By Definition 2, were this observation made at the individual level, holding beliefs about the outcome fixed, it would indicate the presence of taste-based discrimination. A motorist who lies on the indifference point is called *marginal*. As analysts only posses information at the level of average outcomes, one version of the outcome test has sought to provide conditions under which those averages—or in this running example, hit rates—are informative about the double standard. This variety of the test is the subject of this section. Another version of the test targets the marginal motorist directly, by appealing to a large econometric literature on marginal treatment effects, which require the use of instrumental variables (Heckman and Vytlacil, 2007). I address this type of test in Section 4.

In what follows, I show how the framework described in Section 2 offers a general, complete and intuitive causal characterisation of the inferential difficulties that arise from using hit rates to deduce discrimination, as defined by Definition 1. To begin, let me define the hit rate disparity

$$\Delta \coloneqq \mathbb{E}[Y \mid D = d_1, R = r_1] - \mathbb{E}[Y \mid D = d_1, R = r_0]. \tag{7}$$

This quantity compares the success of the officer's searches among minorities to that of White drivers. The outcome test aims to infer discrimination against minorities from the empirical observation that $\Delta < 0$, reflecting the simple Beckerian intuition of the test: if officers find contraband less often in minority vehicles, then the standard applied to them for search was lower. This inference is not always valid. To see why, first define the principal strata of D by $\text{AS} \coloneqq \{\omega \in \Omega : D(r_1)(\omega) = d_1, D(r_0)(\omega) = d_1\}, \text{PF} \coloneqq \{\omega \in \Omega : D(r_1)(\omega) = d_1, D(r_0)(\omega) = d_0\}$, $\text{AW} \coloneqq \{\omega \in \Omega : D(r_1)(\omega) = d_0, D(r_0)(\omega) = d_1\}$, and $\text{NS} \coloneqq \{\omega \in \Omega : D(r_1)(\omega) = d_0, D(r_0)(\omega) = d_0\}$ as the always-searched, profiled (anti-minority) searches, anti-White-searches, and never-searched respectively. Note that as the hit rate conditions on

a search taking place, the group NS bears no relevance here. I have the following proposition.

Proposition 2 *The hit rate decomposes as* $\Delta = (1) + (2) - (3)$*, where*

$$(1) \quad \mathbb{E}[Y \mid \text{AS}, R = r_1] - \mathbb{E}[Y \mid \text{AS}, R = r_0],$$

$$(2) \quad \Pr(\text{PF} \mid D = d_1, R = r_1)\big\{\mathbb{E}[Y \mid \text{PF}, R = r_1] - \mathbb{E}[Y \mid \text{AS}, R = r_1]\big\} = \frac{\text{Cov}(Y(d_1),\, D(r_1) - D(r_0) \mid D = d_1, R = r_1)}{\Pr(\text{AS} \mid D = d_1, R = r_1)},$$

$$(3) \quad \Pr(\text{AW} \mid D = d_1, R = r_0)\big\{\mathbb{E}[Y \mid \text{AW}, R = r_0] - \mathbb{E}[Y \mid \text{AS}, R = r_0]\big\} = -\frac{\text{Cov}(Y(d_1),\, D(r_1) - D(r_0) \mid D = d_1, R = r_0)}{\Pr(\text{AS} \mid D = d_1, R = r_0)}.$$

Proof. See Appendix A.2. ■

As discussed earlier in this section, naïve intuition would suggest that a large, negative Δ implies the presence of discrimination against minorities. It is well-understood that this intuition breaks down due to what is called the inframarginality problem (e.g. Ayres, 2002). Briefly, it is a type of identification failure due to omitted variable bias: Δ averages over characteristics V away from the target marginal motorists, meaning that Δ compares very different kinds of drivers. For example, even if officers apply exactly the same search standard to both races, $\Delta > 0$ may arise simply because the minority motorists who are searched tend to have higher underlying contraband risk than the White motorists who are searched. I will discuss how this bias arises in my analysis below. For additional expositional treatments of inframarginality bias see (Ayres, 2002; Simoiu et al., 2017).

Proposition 2 goes beyond standard discussions of the inframarginality problem to outline mechanisms by which an inference of discrimination generally does not follow. I discuss each term in turn. I call (1) *pure inframarginality*. More precisely, the always-searched units may be regarded as those for whom officers use little discretion in their decision to search—think of vehicles emanating a strong odor associated with the sought contraband, for instance. Term (1) is nonzero if different racial groups among the always-searched are

found to possess contraband at different rates when searched. Within a given context of a stop ω, e.g. location, time of day, etc., as argued by Ayres (2002, p. 137), a story motivating a disparity in (1) would be hard to maintain. I discuss this assumption in detail in Section 5.1 and the empirical application along with examples. However, the impact of unobservables in V could in principle render this term nonzero.

Next, consider (2), which I call the *productivity of racial profiling*. The pre-multiplying weight, $\Pr(\text{PF} \mid D = d_1, R = r_1) = \Pr(D(r_0) = d_0 \mid D = d_1, R = r_1)$ is known as the probability of necessity in the causal literature (Pearl, 1999; Tian and Pearl, 2000), or sometimes, the attributable effect (Rosenbaum, 2005). For legal purposes, this quantity is desirable as a measure of discrimination due to its tight connection to legal stature: for stops of minorities who were searched, it directly quantifies the likelihood of the search (not) occurring were the driver White. Therefore, term (2) is only active in the event that discrimination is present, i.e. $\Pr(D(r_0) = d_0 \mid D = d_1, R = r_1) > 0$. The difference in braces is the revealed disparity in contraband discovery due to racial profiling: this term is positive if in encounters where the officer would discriminate, she is more successful at discovering contraband than in encounters where she would not. Overall, term (2) is the component of the hit rate that is explained by an interaction between discrimination and actual risk of contraband. Another way to understand this term is provided by the covariance term on the right-hand side. This alternative representation implies that that a large and negative hit rate disparity comes from a notion that conditional on deciding to search a vehicle, discovery of guilt is negatively correlated with discriminatory practices, or, that racial profiling is *unproductive* for producing contraband, hence its name. Terms (2) and (3, *productivity of anti-White profiling*) are mirror images of each other and their balance is informative for the sign of Δ.

Proposition 2 reveals why the sign of Δ is not informative of discrimination. First suppose that discrimination is absent, or graphically, we remove all emanating arrows from R. This implies that the profiled and anti-White searched groups are empty, eliminating terms (2)

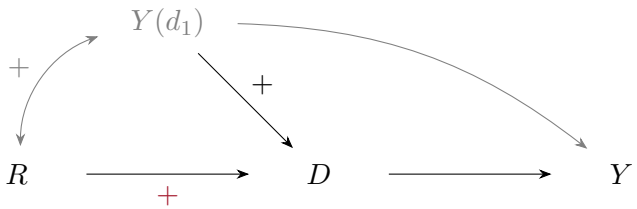


Figure 2: **Conditioning on a collider.** Under officer clairvoyance, $B = Y(d_1)$, it suffices to take the confounder $V = Y(d_1)$. The goal of the test is to detect if R has a positive effect on D, indicated by the red +. Hit rates condition on the collider D, forcing the analyst to wrestle with several interaction effects to infer discrimination. Conditions (i) and (ii) of Proposition 3 is reflected graphically by the '+' signs on the relevant edges using notation from the signed graphs literature (VanderWeele and Robins, 2010).

and (3). However, the path $R \leftrightarrow V \rightarrow Y$ remains open, meaning that (1) is potentially nonzero. This reflects the prevailing wisdom that differences in characteristics relevant for the outcome away from the margin lead to bias in the test.

The above argument reasons that an analyst could incorrectly infer discrimination in its absence. Equally, she may make the wrong inference in its presence the test can produce an erroneous implication of its direction. To sharpen the point, impose some favorable conditions. Suppose there is no anti-White bias, so that $D(r)$ is monotonic in r. Further, suppose that pure inframarginality does not exist, so (1) is zero. Then it follows from Proposition 2 that $\Delta = (2)$. Assume discrimination is present, so that $\Pr(D(r_0) = d_0 \mid D = d_1, R = r_1) > 0$. If the term in braces is also positive, then $\Delta > 0$ failing to indicate discrimination. A story that could motivate this observation is uncommon but not impossible: suppose the jurisdiction of interest is a diverse neighborhood in a city that also happens to be the territory of a drug cartel whose membership is exclusionary based on a racial criterion (e.g. the Latin Kings, where only Hispanic residents may join). Then, it could be possible that using race as a cue for drug trafficking is productive for discovering contraband. This phenomenon leads the hit rate to show the wrong sign. Equally, the discrimination could be present but $\Delta = 0$ because race is exactly productive in producing contraband in a manner which balances the non-discretionary cases.

Proposition 2 gives a new characterization of the outcome test which argues that hit rates

are uninformative for detecting the presence of discrimination. Importantly, the discussion above indicates that the inferential complexity of the test is due to the presence of many potential interactive phenomena in the manifestation of the hit rate disparity. I will now provide deeper for intuition for why these complexities arise by showing how it is a pernicious consequence of conditioning on a collider. Again, we will assume favorable conditions for the analyst as restrictions on the model highlight the pervasive nature of the issue. Write $V = (B(r_0), B(r_1), Y(d_1))$ as a sufficient low-dimensional representation of the non-racial features of the stop. This trick is sometimes used in causal arguments to reason about confounding when confounders are high-dimensional. The idea is that ignorability conditions become trivial when the unobserved confounders are the relevant principal strata.[4] Now, suppose the officer is *clairvoyant*, so that $B = Y(d_1)$. In other words, the officer can perfect predict the future in that she knows whether or not she will find contraband if she searches the vehicle. It follows that we can simply take $V = Y(d_1)$, i.e. the found guilt of the motorist, see Figure 2. I have the following result.

Proposition 3 *Suppose officers are clairvoyant. If the analyst observes* $\text{Cov}(R, Y \mid D = d_1) < 0$*, then she can infer the presence of racial bias in the sense of Definition 1, if the following three conditions hold:* (i) $\text{Cov}(R, Y(d_1)) > 0$*,* (ii) D *is monotonic in* $Y(d_1)$*, and* (iii) $\Pr(D = d_1 \mid R = r_1, Y(d_1) = 1) - \Pr(D = d_1 \mid R = r_1, Y(d_1) = 0) > \Pr(D = d_1 \mid R = r_0, Y(d_1) = 1) - \Pr(D = d_1 \mid R = r_0, Y(d_1) = 0)$.

Proof. See Appendix A.3. ■

To build intuition for this result, consider for a null hypothesis that racial bias is absent, $R \nrightarrow D$. Minority motorists may nevertheless be searched more frequently because they are more likely to be guilty, this is condition (i). Guilt increases the probability of search, this is condition (ii). Under the null, therefore, a positive association between race and search therefore need not indicate discrimination. However, under this null, search depends on guilt but not additionally on race. Thus, searching motorists under these conditions cannot

[4]For examples, see Tchetgen Tchetgen (2014); Ding et al. (2017); Duarte et al. (2024).

reverse the positive association between race and guilt. Consequently, were an analyst to observe

$$\text{Cov}(R, Y(d_1)) > 0 \qquad \text{but} \qquad \text{Cov}(R, Y(d_1) \mid D = d_1) < 0 \tag{8}$$

she must reject the null hypothesis that race has no direct impact on the search decision (absent unmeasured confounding, which holds in this stylized example).

The reversal in (8) does not yet determine the direction of discrimination. Once the null that race does not affect search has been rejected, search becomes a common effect of race and guilt, and conditioning on searched motorists can itself induce an association between the two. If minority status raises the propensity to search, then minority motorists may be searched on weaker evidence of guilt, while White motorists must exhibit stronger evidence before being searched. Among those who are searched, minority motorists will therefore tend to be less guilty on average. In this sense, discriminatory search behavior can generate a negative association between race and guilt among searched motorists.

The reverse direction requires more care. Observing that searched minority motorists are less likely to be guilty does not necessarily show that minority status increased their probability of search. The same pattern could arise if officers were simply much better at distinguishing guilty from innocent White motorists than guilty from innocent minority motorists. For example, suppose officers rarely search innocent motorists of either race, but are very likely to search guilty White motorists and only moderately likely to search guilty minority motorists. Minority motorists could then have a lower hit rate among those searched even if, at every level of guilt, they were actually less likely to be searched than White motorists. The lower minority hit rate would reflect weaker sorting on guilt rather than a greater propensity to search minorities.

Condition (iii) in Proposition 3 rules out this alternative by requiring officers to be more sensitive to guilt when deciding whether to search minority motorists as when deciding whether to search White motorists. Under this restriction, a lower hit rate among searched minority motorists can no longer be explained by officers sorting less effectively on guilt

for minorities. Combined with the earlier conclusion that race affects the search decision, the negative hit rate disparity can then be interpreted as evidence of discrimination against minorities.

Together, the proposition shows that inferring the sign of discrimination is inseparable from an assumption about this interaction: a conclusion of racial bias requires the analyst to assert that race amplifies the officer's reaction to signals of guilt. Indeed, this is a specific manifestation of *collider bias*, where the decision variable D is a descendant of the collider belief, see Figure 1. Indeed, it is well known that conditioning on a common effect distorts the relationship between its causes, even if independent (Pearl, 2009; Elwert and Winship, 2014; Hernán et al., 2004). The direction of this distortion depends on the sign of the effect of the parents on the child (Ding et al., 2017; Nguyen et al., 2019). Proposition 3 provides an inverse result, giving sufficient conditions for the relationship between R and $Y(d_1)$, in addition to their joint impact on D, in order to infer the effect of R on D.

The purpose of Proposition 3 is to use the general treatment of the outcome test my framework provides to offer a blueprint and a warning for any empirical study which aims to infer racial discrimination from hit rates. An analyst first wrestles with an interaction in order to deduce discrimination, even in the presence of inframarginality bias as typically formulated. Notice that this takeaway of Proposition 3 persists even when officer beliefs are perfect.

To exemplify the importance of this proposition, it is instructive to examine exactly how such assumptions have arisen in published literature. I will revisit two seminal works in econometrics on the outcome test which derive game-theoretic equilibria permitting an inference from Δ, and a more recent work which uses Proposition 3 directly, but unknowingly.

3.1 Knowles et al. (2001) Revisited

Game-theoretic reasoning has been used in economics and political science to tie marginal motorists to average hit rates. These models are attractive as (i) their use in economic

reasoning is widespread and (ii) they control individual behavior in a manner that is mathematically tractable. I will discuss two of these models and show how they can be considered special cases of the causal model I present in Section 2 while also implicitly restricting the chaotic effects of conditioning on a collider.

I start with Knowles et al. (2001), henceforth KPT. I will describe their model briefly, adapting their notation to this article. As a consequence of an assumed utility maximizing criterion, officers are assumed to search a vehicle of type V with driver of race R if their belief[5] $B(r, v)$ about the discovery of contraband exceeds their threshold $t(r)$, and will not search the vehicle if it does not. Note that in this model all officers are assumed to possess the same threshold $t(r)$. That is, $t(r)$ does not vary with U_D. If their belief exactly equals their threshold, the officer is assumed to be 'willing to randomize' over whether or not to search the vehicle. In other words, the structural equation for D is such that

$$f_D(r, b, u_d) = \mathbf{1}\{b > t(r)\} + g(u_d)\mathbf{1}\{b = t(r)\} \tag{9}$$

for some $g(u_d) \in \{0, 1\}$ such that $\Pr(g(U_D) = 1 \mid B = t(r)) \in (0, 1)$. This is a behavioral assumption on the existence of officer indifference and their response to it. This assumption imposes no functional restriction on the full data law, however, as the KPT model progresses, the target of inference will center around the marginal units for whom $b = t(r)$. While arguably innocuous a-priori, I show that the authors' reasoning will lead to observable implications of this idea which amount to eliminating heterogeneity. In equilibrium, an officer is said to exhibit taste-base discrimination if $t(r)$ is not constant in r. According to KPT, statistical discrimination arises when, if a taste for discrimination is absent, one finds that $\Pr(D = d_1 \mid R = r_1) \neq \Pr(D = d_1 \mid R = r_0)$, I return to this definition shortly.

Motorists are assumed to possess some payoff $-j(V, R)$ if not searched, and $v(V, R)$ if

[5]KPT do not work primitively with the officer's belief, in their model this quantity is replaced with the *actual* probability of guilt. I write it in this manner to (i) tie their thinking to the causal model, (ii) emphasize how their results rely on calibrated beliefs.

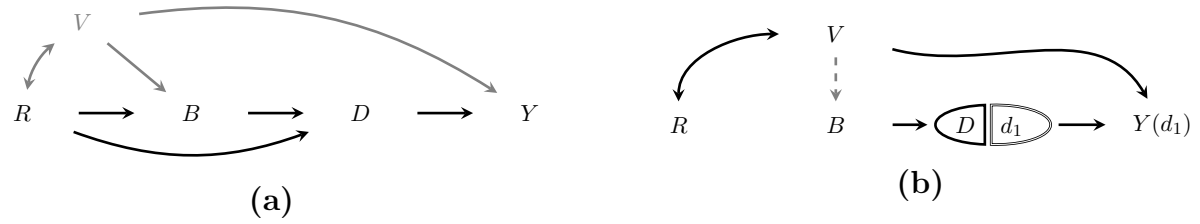


Figure 3: **A Single World Intervention Graph describing the data-generating process in equilibrium.** Panel (a) depicts the full causal model for convenience of comparison. Panel (b) gives a Single World Intervention Graph describing the data-generating process in equilibrium. The hypothesis of no taste-based discrimination is reflected by the absence of an edge $R \to D$. The model implies that all *searched* individuals are marginal, which additionally removes the edge $V \to B$, removing all heterogeneity in beliefs.

searched, where both $v, j > 0$. Therefore, their expected payoff from carrying contraband is

$$\Pr(D = d_1 \mid V, R)[-j(V, R)] + \Pr(D = d_0 \mid V, R)[v(V, R)]. \tag{10}$$

A motorist is assumed to choose to carry contraband when this term is positive, and is willing to randomize when this term is zero. The authors show that an equilibrium is reached when both motorists and officers are indifferent to their corresponding choices. That is, when motorists are willing to randomize, officers search with intensity $v(V, R)/(v(V, R) + j(V, R))$, which is between zero and one, so officers are also willing to randomize. Such an officer must have that their belief exactly equals their threshold, among those they chose to search.[6] Therefore, under the hypothesis that taste-based discrimination does not exist, it follows that $B(r_1, V) = B(r_0, V) = t$ whenever $D = d_1$. From this argument it follows that $\mathbb{E}[Y \mid D = d_1, R = r] = \mathbb{E}_{V|D=d_1,R=r}\mathbb{E}[Y \mid D = d_1, R = r, V] = \mathbb{E}_{V|D=d_1,R=r}\mathbb{E}[Y \mid B(r, V) = t, D = d_1, R = r, V] = t$ if $B(r, V) = \mathbb{E}[Y(d_1) \mid V, R = r]$, that is, officer beliefs are calibrated.[7] This model brings forward the testable implication that the hit rate is constant in r.

In Figure 3, I draw a Single World Intervention Graph (Richardson and Robins, 2013) which describes the wrangling of the causal model induced by the equilibrium above, under

[6] To be clear, officer beliefs need not be indifferent over all motorists, only those they choose to search. Therefore, even in equilibrium, officers may search drivers of different racial groups at different intensities.

[7] I examine the general use of assumptions on the calibration of beliefs in Section 4.

the hypothesis of no taste-based discrimination. Figure 3 exemplifies the caution offered by Proposition 3. Indeed, the equilibrium of this game implies that within the searched, all units are marginal, meaning that any heterogeneity in beliefs is assumed away, and trivially, the inframarginality problem is solved. Moreover, by hypothesis, R is assumed to have no direct effect on D, but, additionally, a property of the equilibrium above is that R did not affect beliefs about contraband among the searched. Graphically, this properties equates to the removal of any emanating arrows from R, and any incident arrows to B, the equilibrium removes any interaction between R and V to influence beliefs and subsequently search decisions. The nuance generated by conditioning on collider D is completely eliminated.

Connecting this model to the causal framework proposed in this paper, the assumed absence of heterogeneity allows us to rapidly arrive at the same result via the principal strata analysis of Proposition 2. In the KPT model, among the searched, all units are marginal. Therefore, there is no distinction between the principal strata, among the group of motorists where $D = d_1$. Since it is assumed among this group that the officer's beliefs about contraband risk are calibrated, and importantly, homogeneous in V, the contraband risk across the strata is identical (within each racial group). Thus, each term in Proposition 2 collapses to zero.

Game-theoretic notion	**Causal notion**
Marginal motorist	Existence of V such that $B(r, V) = t(r)$
"Willingness to randomize"	Functional restriction on f_D and U_D
Taste-based discrimination: $t(r_1) \neq t(r_0)$	nonzero NDE on individual level
Statistical discrimination: $p(d_1 \mid r_1) \neq p(d_1 \mid r_0)$ if $t(r_1) = t(r_0)$	Confounding in the absence of $R \to D$ effect
"In equilibrium"	Beliefs are homogeneous in V and R

Table 1: A translation glossary between some causal and game-theoretic notions.

Objections to the KPT model have appeared in the literature (e.g. Dharmapala and Ross, 2003; Engel, 2008), chiefly on the grounds of their extreme behavioral assumptions, e.g. how can motorists "respond" to the probability an officer will search them? Additionally,

I highlight that KPT require that officer beliefs correspond to actual contraband risk, since otherwise the hit rates are not guaranteed to equalize, rendering the homogeneity they exploit fruitless, as beliefs are not measurable.[8] Also, KPT argue that a failure to reject equal hit rates but a finding that search rates differ by race is evidence of statistical discrimination. However, the graph underlying their world in equilibrium shows why this inference may not be valid, as, here, the restoration of the arrows $R \to B$ and $V \to B$ mean that the disparity $\Pr(D = d_1 \mid R = r_1) \neq \Pr(D = d_1 \mid R = r_0)$ could equally have been produced by simple confounding via the backdoor $R \leftrightarrow V \to B \to D$, as not all of V could have been measured by the authors in their empirical study to block this path. This line of reasoning demonstrates a point I raised in Section 2.1, that when statistical discrimination is sought as a competing explanation for taste-based discrimination, it can easily be confused with confounding. Consequently, KPT's seminal formulation of the outcome test is in fact incapable of accomplishing its specified goals.

Until now, a formal causal treatment of the flaws of the KPT model has not been presented. In Table 1, I give a translation glossary of some game-theoretic notions as they are applied in this example, tying them to their corresponding causal meaning. I find drawing these connections informative, as they serve as a caution against elegant mathematical reasoning when it conflicts with reasonable predictions of the data generating process.

3.2 Anwar and Fang (2006) Revisited

Anwar and Fang (2006) describe an equilibrium which is conceptually similar to KPT. In response to many of the critiques of the KPT model, their primary innovation is to allow an exogenous instrument—officer race—into the decision to permit inference from hit rates in the presence of inframarginality bias. In my proposed causal framework, officer race can simply be extracted as a component of officer type U_D. Let officer race be described by the binary variable Q; denote q_0 a White officer and q_1 a minority officer. Like KPT, Anwar and

[8] The authors do not acknowledge this restriction—that officers respond to the true probability of contraband discovery given (R, V) is taken as primitive.

Fang assume that that officer beliefs are calibrated, in that they exactly equal the true risk of motorists conditional on V and R. Importantly, this assumption implies further that beliefs do not vary by the race of the officer. These authors also assume homogeneity in officer standards, however, this statement is now conditional on the officer's race, Q. I now proceed to provide a very simple explanation of their result with none of the additional parametric structure underlying their model, nor any game-theoretic reasoning whatsoever.

Anwar and Fang proceed by examining the marginal motorist in equilibrium, following KPT. However, unlike KPT, their model does not predict homogeneity across marginal motorists because in principle thresholds differ by officer race. One way to examine how variation in Q generates useful variation in the threshold applied to otherwise identical motorists is to think about the population of motorists as a mixture of principal strata in the effect of Q on D conditional on $R = r$. These are the motorists who would be searched by all officers (always-searched), and those who would only be searched by stricter officers ('compliers'), those who would only be searched by more lenient officers ('defiers'), and those who would never be searched, see Figures 4(a) and (b).[9]

This heterogeneity restores the role of inframarginal individuals. Enforce the null of no discrimination, removing all emanating arrows from R. Inspect Figures 4(b) and (c). For the sake of hypothesis, suppose that White (q_0) officers, are more stringent than minority officers. This is a monotonicity restriction on the effect $Q \to D$, therefore, it is impossible for a White officer to search a motorist a minority officer would not, eliminating the 'defier' quadrant of (b). Among the searched vehicles there are clearly no never-searched vehicles, leaving only the 'complier' and always-searched vehicles.

Figure 4(c) shows the relationship between what the beliefs of the officer must be to search a given type of motorist. By hypothesis, $D(q_1) \geq D(q_0)$, and invoking that beliefs do not vary by officer race, it follows that $t(q_1) < t(q_0)$. As beliefs are assumed calibrated, the

[9]I am using the terms complier and defier here for lack of a better choice of words. I remind the reader that these strata are over different types of officer, so the notion of a profiled or anti-White search does not apply here, hence the change in terminology.

location of these strata can be mapped directly to true contraband risk. Crucially, compliers are lower risk than the always-searched. Immediately, one deduces from this that the hit rate of minority officers is smaller than the hit rate of White officers, since minority officers search compliers and the always-searched, but White officers only search the latter. Thus, under the assumption of no taste-based discrimination, for a given motorist race r, the officer type with a higher search rate must exhibit a lower hit rate.

This observation is merely a statement of interaction induced by conditioning on a collider. The decision node D, is a collider on the path $B \rightarrow D \leftarrow Q$. By Proposition 1, stronger beliefs about contraband risk induce a higher likelihood of search. Therefore, whenever the effect of Q on D is positive, conditioning on search induces a negative relationship between B and Q, a property which holds conditional on each $R = r$. The authors, like KPT, assume that officer beliefs are calibrated, such that B is true risk of contraband conditional on (R, V). This then implies that when conditioning on search, if search rates increase in Q, then rates of actual contraband discovery must decrease in Q, restating the result above, however, much more briefly. This testable implication is precisely Proposition 3 in the authors' paper (Anwar and Fang, 2006, p. 136).

Again, this seminal work in the outcome test literature unknowingly unveils the blueprint drawn up by Proposition 3: to derive an inference of racial bias, one must first find evidence in support of an interaction effect created by conditioning on a collider.

3.3 Gaebler and Goel (2025) Revisited

A recent result by Gaebler and Goel (2025) provides a particularly direct application of Proposition 3. Their empirical setting concerns consumer lending. A lender observes information V about an applicant, and the authors define $B(r, v) = \Pr(Y = 1 \mid R = r, V = v)$ to be the applicant's true probability of repaying a loan. Applicants of race r receive a loan if and only if B exceeds a race-specific threshold, $t(r)$, so that a higher threshold represents a more stringent lending standard. There is assumed to be only one decision-maker here, the

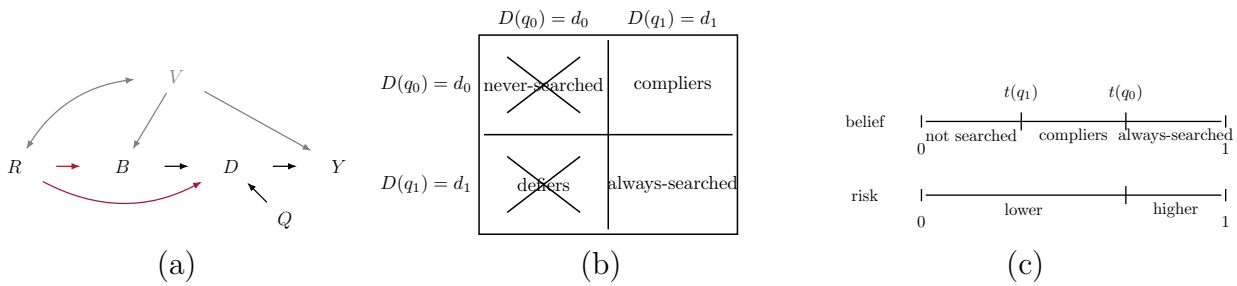


Figure 4: **Illustration of Anwar and Fang test.** (a) The DGP. (b) Principal strata on D in the absence of discrimination. (c) Mapping from beliefs to strata, with corresponding risk, if beliefs are calibrated. Under the null, the red arrows (those emanating from R) in (a) vanish, and suppose for hypothesis that $D(q)$ is monotonic in q, so that among those searched, we have no never-takers or defiers. Equally, the White (q_0) officer's threshold is higher than the minority (q_1) officer's threshold, see (c). Assuming beliefs are calibrated, the contraband risk among minority officer searches is lower than that of White officers.

lender.

Their workhorse assumption is that $\Pr(R = r_1 \mid B = b)$ is monotone in b, which is equivalent to the monotone likelihood ratio property (MLRP) for the race-specific risk distributions. Their main result states that if applicants of race r_1 receive loans less frequently than those of race r_0, but repay their loans more frequently conditional on receiving them, then $t(r_1) > t(r_0)$. Thus, applicants of race r_1 face a more stringent lending standard.

At its core, their result follows the blueprint drawn by Proposition 3. Take the non-decreasing version of the authors' MLRP assumptions such that $\Pr(R = r_1 \mid B = b)$ weakly increases with b. Moreover, because lenders approve applicants with sufficiently high B, D is monotonic in B. These correspond to conditions (i) and (ii) of Proposition 3. Now suppose one observes that applicants from r_1 receive loans less frequently than applicants from r_0, yet repay more frequently among those who are approved. Under the null ($t(r_1) = t(r_0)$), a greater repayment probability of applicants from r_1 can explain a higher approval rate, but cannot explain the opposite pattern in which they receive loans less frequently while repaying more frequently when approved. Therefore, the empirical observation rejects the null of no discrimination.

Without further restriction, however, the observed disparities do not yet determine which

group faces the higher threshold. Applicants from r_1 could, for example, be disproportionately concentrated at both very low and very high repayment probabilities, while applicants from r_0 are concentrated in the middle. In that case, r_1 applicants could face a lower lending threshold and nevertheless be approved less often, simply because so many lie below even that lower threshold. At the same time, the r_1 applicants who do clear the threshold could be concentrated at very high repayment probabilities, producing the higher repayment rate observed among approved borrowers.

The MLRP assumption rules out precisely this kind of crossing in the race-specific repayment distributions. In its non-decreasing form, applicants from r_1 become progressively more prevalent as repayment probability increases, relative to White applicants. Thus, r_1 applicants are ordered toward higher repayment probabilities. If they also faced a weakly lower lending threshold, they would be approved at least as often as r_0 applicants. Their observed lower approval rate therefore rules out this direction of discrimination and implies $t(r_1) > t(r_0)$. The higher repayment rate among approved r_1 applicants is the consequence of that stricter standard.

The reader will notice that the intuition underlying Gaebler and Goel's main theorem closely parallels that of the proof in Proposition 3. The non-decreasing MLRP restriction supplies the analogue of condition (i), while monotonicity of the lending decision in repayment probability supplies condition (ii). But MLRP is considerably stronger than the restriction in condition (i). By imposing a global ordering on the repayment distributions by race, it also rules out the competing distributional configurations discussed above that could otherwise generate the observed disparity among the approved under the opposite ordering of lending thresholds. In this sense, the additional content of MLRP plays the same inferential role as condition (iii) in Proposition 3: each rules out an alternative explanation that would otherwise prevent the sign of discrimination from being recovered. Briefly stated, the MLRP condition, along with monotonicity, is strong enough to enforce all three conditions of Proposition 3 at once.

All three examples in published work highlight why Proposition 3 is important for any application of the outcome test. Hit-rates unleash many interacting mechanisms between the causal parents of the search decision; strictly eliminating, controlling, or leveraging them with strong assumptions is necessary to deduce racial bias. Importantly, the purpose of this proposition is not to argue that these designs do not credibly reveal bias, but rather blueprints what kinds of assumptions will need to be defended in order to do so. I discuss the credibility of these assumptions shortly.

4 The Elusive "Marginal Motorist" Police Are Indifferent to Searching

In Section 3, I examined the use of hit rates to infer racial bias, and found that it is generally a delicate endeavor. I now turn to the second general form of the outcome test. Rather than use assumptions which make average hit rates informative about motorists at the margin, this approach targets the marginal motorist directly. This latter form of the outcome test is called a *marginal outcome test*. The primary advantage of this strategy is that it removes the particular problem of averaging over infra-marginal motorists. In this section, I discuss versions of the outcome test designed to identify thresholds directly.

4.1 Instruments, MTEs and thresholds

One approach uses theory from the widely used marginal treatment effects literature in econometrics (Heckman and Vytlacil, 2007). I will not review this literature here, but show how the relevant estimand may arise from the causal model (2) with an additional component: an instrumental variable.

The appeal of an instrument is that it generates exogenous variation in the decision to search. Rather than learning about guilt by conditioning on $D = d_1$, the analyst compares motorists whose probability of search differs for reasons unrelated to their potential outcomes.

This provides a route around the collider bias discussed earlier. To see how the marginal treatment effect arises in model (2), augment the model with an instrument Z that affects D but has no direct effect on Y. In recent criminal-justice applications, Z is commonly the identity or leniency of a quasi-randomly assigned judge (Arnold et al., 2018, 2022). Under typically invoked IV independence and monotonicity conditions (Vytlacil, 2002), for each race r the decision admits the latent-index representation $D = \mathbf{1}\{\varepsilon_D \leq p_r(Z)\}$, where $\varepsilon_D \mid R = r \sim \text{Unif}(0,1)$ and $p_r(z) = \Pr(D = d_1 \mid R = r, Z = z)$. With sufficiently rich variation in $p_r(Z)$, the local instrumental variables estimand identifies the race-specific marginal treatment effect,

$$\text{MTE}_r(p) = \frac{\partial}{\partial p}\mathbb{E}[Y \mid p_r(Z) = p, R = r] = \mathbb{E}[Y(d_1) \mid \varepsilon_D = p, R = r], \tag{11}$$

because $Y(d_0) = 0$ in the running example. Intuitively, an infinitesimal increase in p induces motorists with $\varepsilon_D = p$ to be searched, so (11) recovers the expected guilt of motorists at that margin of the decision, rather than averaging over all searched motorists. The marginal outcome test then compares these quantities across races at their respective decision margins. Let p_r^* denote the relevant race-r margin and define

$$\Delta_{\text{MTE}} := \text{MTE}_{r_1}(p_{r_1}^*) - \text{MTE}_{r_0}(p_{r_0}^*). \tag{12}$$

This strategy directly addresses inframarginality: the instrument recovers the outcome of units at a margin rather than requiring average outcomes among all searched motorists to represent it. However, what is identified in (11) does not automatically correspond to the subjective point of indifference of any particular decision-maker. This distinction becomes especially important in judge-leniency designs, where the instrument indexes heterogeneous decision-makers, whose thresholds differ from judge to judge. Connecting the MTE identification strategy to a decision-maker's standard therefore additionally requires structure which aligns these standards, for example by conditioning on decision-maker identity (e.g. Arnold

et al., 2018, 2022) or imposing sufficient homogeneity in thresholds across the decision-makers being pooled; examples include the study of KPT and Anwar and Fang considered here, a more recent use of this assumption is found in Marx (2022).

For typical applications of the outcome test, identifying a valid instrument is difficult. Recent applications of the MTE framework have used judge-leniency designs (e.g. Arnold et al., 2018). Such designs require stringent monotonicity restrictions on how judges rank defendants (Vytlacil, 2002). Hull notes that differences in these rankings generally violate this assumption (Hull, 2021), though, this is an observation that has been made more broadly (Frandsen et al., 2023; Cooper et al., 2026a).

Threshold tests. One identification strategy motivated by the outcome test is the threshold test of Simoiu et al. (2017). I mention it here because of its focus, like the MTE framework, on targeting the threshold directly. In its original formulation, the authors specify a parametric Bayesian hierarchical model for the decision-making process, where officers act in response to a latent probability of contraband discovery homogeneous within each race and police department, and all officers within a department are assumed to apply the same standards by race. Thus, heterogeneity in motorists is modeled through the background risk distribution while heterogeneity in officer thresholds is largely removed. The model additionally assumes officers respond directly to the true contraband risk, which is analogous to earlier examples discussed in this article showing where officers are assumed to rely on calibrated beliefs.

Viewed through the framework developed in this article, threshold tests are simply highly structured special cases of model (2). They amount to equipping the variables and disturbances in the causal model with particular distributions and restricting U_D such that the relevant decision-makers share a threshold. They avoid the need for an instrument, and explicitly address inframarginality, but retain several of the other difficulties considered throughout this article that have repeatedly appeared, such as, substantial homogeneity

across decision-makers is imposed, and that beliefs must be calibrated in strict sense.

4.2 Calibrated Beliefs

The discussion thus far has repeatedly relied on the same tether between the decision-maker's subjective beliefs and the outcomes observed by the analyst. It appears in the game-theoretic models of KPT and Anwar and Fang, in the lending model of Gaebler and Goel, and again in marginal outcome and threshold tests. Crucially, this connection lies at the core of the original Beckerian intuition for the test. The 'standard' applied by the decision-maker is a statement about their subjective point of indifference: the belief at which she is just willing to act. The outcome test, however, observes a realised outcome—e.g. whether contraband is actually found. Thus, even if the marginal individual were known exactly, her observed outcome would reveal the decision-maker's threshold only if the belief on which the decision was based corresponds appropriately to objective risk. I refer to this correspondence as calibration, and state it precisely below.

Definition 4 (Calibrated belief) *Fix r and $v \in \text{supp}(V \mid R = r)$. The belief $B(r, v)$ is calibrated if*

$$B(r, v) = \mathbb{E}[Y(d_1) \mid B(r_0, v) = b_0, B(r_1, v) = b_1, R = r],$$

that is, the belief the officer forms under race r coincides with the true conditional probability of contraband discovery given the pair of beliefs (b_0, b_1) she would form of the same motorist.

Definition 4 conditions on the pair of beliefs $(B(r_0, v), B(r_1, v))$ rather than the entirety of V, so that calibration is a statement about the belief the officer actually forms rather than about features of the motorist she may never use. The following proposition shows that a marginal outcome reveals the officer's threshold exactly when her belief is calibrated in this sense, to serve in a similarly cautionary role as Proposition 3.

Proposition 4 (Calibration) *Assume model* (2). *Fix $r \in \{r_0, r_1\}$ and an officer u_d. Let $t(r, u_d)$ denote the search threshold applied by officer u_d to motorists of race r as in* (4).

Define the marginal motorists of race r for officer u_d as those with characteristics belonging to the set $\mathcal{M}_r(u_d) := \{v \in \text{supp}(V \mid R = r) : B(r, v) = t(r, u_d)\}$. Assume that

(i) $\mathcal{M}_r(u_d)$ *is nonempty*

(ii) Marginal motorists are searched with positive probability, i.e.

$\Pr(D = d_1 \mid R = r, U_D = u_d, V \in \mathcal{M}_r(u_d)) > 0.$

(iii) Guilt can be no more informative for the officer than their perception of it,

$D \perp\!\!\!\perp Y(d_1) \mid (B(r_0), B(r_1), R)$.[10]

Then, for any $v \in \mathcal{M}_r(u_d)$, $\mathbb{E}[Y \mid B(r_0, v) = b_0, B(r_1, v) = b_1, D = d_1, R = r, U_D = u_d] = t(r, u_d)$ if and only if $B(r, v)$ is calibrated (Definition 4).

Proof. Fix r, u_d, and $v \in \mathcal{M}_r(u_d)$. By consistency and condition (iii) implied by the DAG in Figure 1, $\mathbb{E}[Y \mid B(r_0, v) = b_0, B(r_1, v) = b_1, D = d_1, R = r, U_D = u_d] = \mathbb{E}[Y(d_1) \mid B(r_0, v) = b_0, B(r_1, v) = b_1, R = r]$. By the definition of $\mathcal{M}_r(u_d)$, $B(r, v) = t(r, u_d)$. It follows immediately that $\mathbb{E}[Y \mid B(r_0, v) = b_0, B(r_1, v) = b_1, D = d_1, R = r, U_D = u_d] = t(r, u_d)$ if and only if $B(r, v) = \mathbb{E}[Y(d_1) \mid B(r_0, v) = b_0, B(r_1, v) = b_1, R = r]$, which by Definition 4 holds if and only if $B(r, v)$ is calibrated. Conditions (i) and (ii) are required so that the probabilities reasoned about above are well-defined. ■

Most methodological development of the outcome test is centered around identifying or making inferences about the marginal motorists, i.e. the target of inference is the officer's evidentiality standard for searching a vehicle. Proposition 4 emphasizes that whenever this is sought, without additional structure laid upon the problem, the analyst is forced to assume that officer beliefs are calibrated.

I make some remarks on the causal properties of this proposition. Calibration in Proposition 4 is shown specific to the officer, which is necessitated by the causal model in fullest

[10]This is what licenses Definition 4 to condition on the belief pair $(B(r_0, v), B(r_1, v))$ in place of v: it guarantees that, once the officer's beliefs about a motorist are fixed, no further information in v helps the officer predict $Y(d_1)$, so the belief pair is a sufficient summary of v for this purpose.

generality. If thresholds vary across officers, then the set of marginal motorists differs by officer. The model studied by Anwar and Fang gives a simple example of how this can happen. There, White and minority officers may apply different thresholds, so a motorist who is marginal for one officer type need not be marginal for the other. Even if the beliefs of every officer are perfectly calibrated, pooling these motorists therefore combines outcomes corresponding to different points of indifference. The resulting average outcome need not equal the threshold of any particular officer. Moreover, if racial groups encounter different mixtures of officers, their pooled marginal outcomes may differ even if officers were racially unbiased. Thus, the above result also emphasizes that inference of a given threshold can be made only by conditioning on the decision-maker, or by imposing and defending sufficient homogeneity in thresholds across officers.

This restriction has appeared in different forms throughout the literature examined above. KPT assume a common race-specific threshold across officers. Anwar and Fang permit thresholds to differ by officer race, but impose homogeneity within officer type. Threshold tests similarly impose common thresholds within the cells over which the model is fit, e.g. race and department. These assumptions are rarely presented as instances of the same problem, but the causal model makes their common role transparent: each restricts heterogeneity in U_D so that a marginal outcome may be attached to a meaningful threshold.

Substantively, available evidence provides little basis for supporting this restriction. Direct assessments of police prediction have found officers' risk classifications to be little better than random at predicting subsequent serious harm (Turner et al., 2019). Experimental evidence also shows that police judgments are systematically affected by racial associations. Graham and Lowery (2004) find that activating the racial category Black caused police officers to judge an otherwise unspecified adolescent as more culpable, more likely to recidivate, and deserving of harsher punishment. Similarly, Eberhardt et al. (2004) show that associations between Blackness and crime altered officers' visual attention and memory, and that officers disproportionately selected Black faces when asked who appeared criminal. Most

directly, recent work by González et al. (2026) elicit crime predictions from experienced police captains while independently varying true crime and its observability. Holding true crime constant across artificial groups, officers were twenty percentage points more likely to attribute greater crime to the group whose crime was more observable. These studies document systematic errors in the predictions upon which officers base their actions. Last, given the systematic nature of these errors, there is no guarantee this problem would be mitigated by averaging across officers.

A more subtle element of Proposition 4 is the assumption that $\mathcal{M}_r(u_d)$, the set of motorists of race r whom the officer is indifferent to searching, is nonempty. Canay et al. (2024) acknowledge that this assumption is required but dismiss it as a weak regularity condition. It is not clear to me that this assumption can so trivially be made. The existence of these motorists is typically invoked for mathematical convenience: provided V is continuously distributed and $B(r, \cdot)$ varies smoothly over its support, some realized motorist will cross the officer's threshold exactly, in much the same way that identification of the marginal treatment effect at a point requires the propensity score to attain that point in the data (Heckman and Vytlacil, 2007). When this support condition fails, as it generically does once the relevant variation is discrete or bounded, the standard response in that literature is to either bound the target parameter or extrapolate under explicitly stated functional restrictions (Brinch et al., 2017; Mogstad et al., 2018). No comparable argument has been offered here, nor elsewhere in the outcome-test literature, where the existence of a marginal motorist is treated as a modeling primitive.

However, the difficulty runs deeper still. Exact indifference presupposes itself a coherent feature of an officer's cognition: that some officer, faced with some motorist, forms a belief matching her threshold to arbitrary precision, and, having done so, behaves as though willing to randomize. Such behavior motivated Luce's theory of utility discrimination, which rejects this premise on the grounds that human comparative judgment does not admit this kind of precision (Luce, 1956; McFadden, 1974). Absent an argument for why officers' search

decisions should instead obey a sharp threshold rule realized exactly by some motorist, treating $\mathcal{M}_r(u_d) \neq \emptyset$ as weak regularity understates what is actually being assumed.

Nearly all major empirical demonstrations of the hit rate or marginal outcome tests have connected beliefs and objective outcomes by assuming, explicitly or implicitly, that the decision-maker's beliefs are calibrated, and that this exact point of indifference exists, as noted by Hull (2021) and Canay et al. (2024). Notice that this was the case in all three major published works examined in the previous sections. The purpose Proposition 4 serves is therefore similar to that of Proposition 3. It does not imply that an outcome test cannot credibly reveal discrimination. Rather, it makes explicit an assumption whose credibility must be defended before the observed disparity can be given its usual interpretation. Building on the work of Bohren et al. (2025), Proposition 4 is especially consequential for attempts to distinguish taste-based from statistical discrimination, it adds to the formalization of the observation that traditional approaches to separate the two cannot stand if beliefs are miscalibrated.

4.3 Marginal Standards are not Causal Estimands

Suppose, in an ideal world, that all of the preceding difficulties discussed in this article were overcome. Tests based on hit rates in Section 3 aim to infer the existence of differential standards. Marginal outcome and threshold tests aim to recover the magnitude of those standards under their corresponding assumptions. Even if credibly identified, in neither case does the resulting threshold disparity quantify how frequently discriminatory decisions are made.[11] This exposes an important feature of much of the outcome test literature: nearly all such studies do not target a causal estimand.

Undesirably, differences in thresholds are not comparable across different settings study-

[11] In recent work, in the context of bail decisions, Arnold et al. (2022) specify their target of interest as the disparity in the release rate among individuals with the same misconduct potential. This choice marks a welcome departure from the literature in that it specifies a non-threshold based estimand, however, identification still proceeds via a marginal treatment effects approach based on a judge-leniency design. Moreover, this estimand is not causal—it measures disparate impact, as opposed to disparate treatment, acknowledged by the authors—and so does not identify discrimination as defined by Definition 1.

ing an identical empirical question, due to changes in background variables V. The same threshold disparity could correspond to very different proportions of individuals who experienced discriminatory outcomes. Communicating such results to policymakers, legal advocates, and other stakeholders is difficult, often requiring additional back-of-the-envelope calculations to translate threshold differences into more interpretable measures of discriminatory treatment. The specification of a target causal estimand of interest would eliminate all these concerns.

In Section 2.1 I provide two new definitions of taste-based and statistical discrimination that may in principle be used to quantify the extent of each type of discrimination, the identification of which I discuss in Section 7. In the next section 5, I begin to outline how a reimagining of the intuition behind the outcome test permits a new class of identification strategies for any causal estimand of interest, avoiding the marginal threshold as a target of inference. I will focus in particular on the probability of necessity, which is a quantity of growing interest in empirical legal work and is intimately connected to the context of stops and searches studied here.

5 Reimagining the Core Intuition of the Outcome Test

Table 2 summarizes the identification strategies and assumptions underlying prominent uses of the outcome test. Each motivates their approach by the traditional Beckerian motivation for the test, focusing on unobserved thresholds defining marginal motorists, leading to the identification issues discussed above.

In this section, I offer a different motivation for the outcome test and, with it, a new class of identification strategies for detecting discrimination. Rather than attempting to recover an unobserved marginal threshold, I use the hit outcome as a proxy for the circumstances that legitimately induce a search. This perspective will motivate a policy-relevant causal estimand closely connected to the hit rate studied in Section 3, while avoiding many of the

Published work	Identification strategy	Principal assumptions
Knowles et al. (2001)	Game-theoretic equilibrium makes all searched motorists marginal	Officer has calibrated beliefs; no heterogeneity in officer thresholds; no heterogeneity in motorists among the searched
Anwar and Fang (2006)	Instrumental variable (officer race)	Calibrated beliefs; beliefs do not vary by officer race; common thresholds within officer race
Gaebler and Goel (2025)	Analogue of Proposition 3	Single decision maker, calibrated beliefs, MLRP by race in repayment probability
Arnold et al. (2018, 2022)	Judge-leniency design	IV assumptions; monotonicity in judge leniency; sufficient variation in the propensity score; calibrated beliefs
Simoiu et al. (2017)	Impose fully parametric model	Completely parametric; strong homogeneity assumptions on the thresholds; calibrated beliefs

Table 2: **Prominent implementations of the outcome test and their identifying restrictions.** Although these approaches differ substantially in how they use observed outcomes, each seeks to recover or make inference about a latent decision standard by placing additional structure on the decision-making process, the distribution of underlying risk, or both.

inferential complexities created by targeting the marginal motorist.

I continue with the study of police searches for contraband. Table 3 introduces the stylized encounters I will use to guide the exposition. They separate why a vehicle would be searched from whether the search would recover contraband, allowing the exposition to track both the motorist's counterfactual search response to race and the productivity of the search. Consider a first example.

Example 1 *Suppose that, in a fictional jurisdiction, there are two types of context V, where $V = 1$ indicates that a strong odor associated with contraband becomes apparent when the officer approaches a vehicle, while $V = 0$ indicates that the officer instead perceives furtive, suspicious movement of the driver. Call searches prompted by the former* fragrant searches *and those prompted only by the latter* furtive searches*. The odor is informative of the presence of contraband, but not perfectly so: were any vehicle searched, contraband would be*

Type	Description	Search response	Role
Fragrant	A strong odor associated with contraband becomes apparent when the officer approaches the vehicle, making recovery relatively likely.	Searched regardless of the driver's perceived race.	Always-searched and comparatively productive.
Furtive (movement)	Upon approaching the vehicle, the officer interprets a purportedly furtive movement as nervousness, a weaker and more subjective indication of contraband.	Searched as minority, not as White.	Profiled search.
Courier	A White motorist perceived as "out of place" is in fact transporting contraband.	Searched as White but not as minority.	Productive anti-White search.
Commuter	A White motorist perceived as "out of place" is merely passing through and carries no contraband.	Searched as White but not as minority.	Unproductive anti-White search.
Inventory	The vehicle is inventoried following an administrative impound	Searched regardless of the driver's perceived race.	Always-searched and comparatively unproductive.

Table 3: **Stylized search types used in the exposition.** The first column gives the labels used to describe the principal strata and the two sensitivity parameters. The motivation for the "furtive" stratum originates in claims made by officers that minorities appear more nervous when approached by police, which is then interpreted as suspicious behavior.

found more often following a fragrant search than following a furtive search,

$$\Pr(Y(d_1) = 1 \mid V = 1) > \Pr(Y(d_1) = 1 \mid V = 0).$$

Now suppose officers search White motorists if and only if $V = 1$*. If these officers do not discriminate against minority motorists, the same rule applies to them. Both the White and minority searched populations would then consist entirely of fragrant searches, and their hit*

rates would be equal:

$$\Pr(Y = 1 \mid D = d_1, R = r_1) = \Pr(Y = 1 \mid D = d_1, R = r_0) = \Pr(Y(d_1) = 1 \mid V = 1).$$

Suppose instead that the analyst observes a lower hit rate among searched minority motorists. Because fragrant searches are equally productive across races, the difference cannot arise if the minority searched population consists only of these motorists. Some minority searches must instead be furtive searches. But officers do not search White motorists under the same circumstances. For these searches, therefore, $D(r_0) = 0$ *but* $D(r_1) = 1$*: the search was attributable to race.*

Here, the use of $Y(d_1)$*, rather than the observed outcome* Y*, is important. The observed outcome is itself partly determined by the search decision: officers can discover contraband only in vehicles they choose to search. If race changes which vehicles are searched, it also changes which vehicles have an opportunity to produce a hit. In that sense,* Y *already reflects the discrimination we are trying to detect. By contrast,* $Y(d_1)$ *reveals what would be found were every vehicle searched, holding the search decision fixed. It provides a noisy trace of whether the underlying circumstances resemble a relatively productive fragrant search or a relatively unproductive furtive search, without simultaneously reflecting whether race caused the officer to look.*

Example 1 serves two purposes: (i) it illustrates how $Y(d_1)$ functions as a negative control outcome for the decision, D, and (ii) simply illustrates how a restriction on the relationship between $Y(d_1)$ and V leads to a hit rate disparity that is informative about discrimination. I next elaborate on each of these points.

Consider (i). Interestingly, by drawing the Single World Intervention Graph (Richardson and Robins, 2013) for the data-generating process under an intervention on the decision to search, one indeed observes that $Y(d_1)$, the discovery of contraband if searched, is precisely a negative control outcome (NOC) (Kuroki and Pearl, 2014; Tchetgen Tchetgen et al., 2024;

Knox et al., 2022) for D in the pursuit of the effect of R on D, i.e. discrimination, see Figure 5. Alternatively expressed, one can think of $Y(d_1)$ as a (racially) non-differential proxy for V. This observation graphically represents the intuition described in Example 1.

This perspective suggests a new interpretation of the outcome test. As a proxy for V, the success of the decision is informative about the circumstances that drive the decision to search but is uncontaminated by discrimination in that decision, because intervening on D severs the path $D \to Y$. The core idea is to use $Y(d_1)$ to learn how the non-racial circumstances represented among searched motorists differ by race, and thereby infer whether race changes which kinds of motorists officers choose to search.

Consider (ii). Example 1 uses principal stratification help to elucidate the required intuition, and I will continue to do so in order to set the stage for the identification approach which ensues. Momentarily suppressing beliefs and repeating the argument made in Section 2, we can take without loss $V = (D(r_0), D(r_1), Y(d_1))$. As in Section 3, we work with the four principal strata are the never-search motorists, NS, the always-search motorists, AS, the profiled search motorists, PF, and the anti-White search motorists, AW. Remark that this trick maintains that $Y(d_1)$ is correlated with $(D(r_0), D(r_1))$.

This representation distinguishes the two pieces of information that motivate the outcome test. The principal stratum tells whether the search was discriminatory, while $Y(d_1)$ gives the objective merit of that decision. The proxy $Y(d_1)$ is useful because it is informative about differences between these groups. For instance, one might expect that fragrant searches turn up contraband much more frequently than searches of vehicles officers chose to ignore regardless of race, i.e. $\Pr(Y(d_1) = 1 \mid \text{AS}) \gg \Pr(Y(d_1) = 1 \mid \text{NS})$. Consider the strata of discriminatory searches, PF and AW, whose prevalence is of primary interest. Their position relative to AS and NS in the distribution of $Y(d_1)$ is informative about the evidentiary standard associated with the racial component of the search decision. For example, if PF motorists tend to have lower $Y(d_1)$ than AS motorists, race is bringing into the searched population motorists with weaker objective evidence of guilt than those who

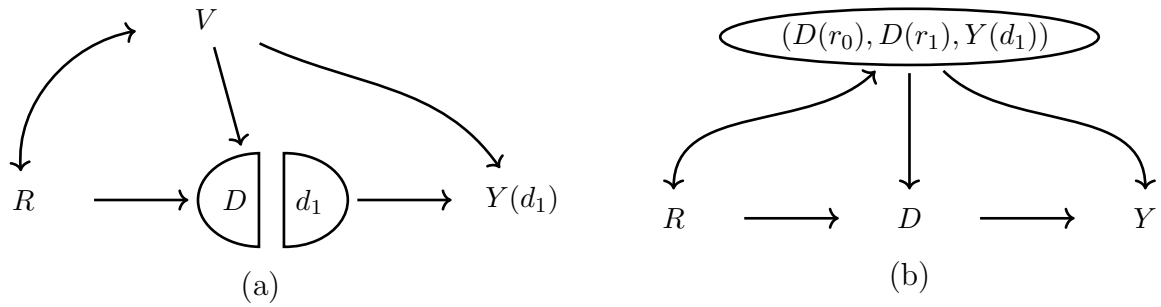


Figure 5: **A new motivation for the outcome test.** Panel (a): intervening on search, a SWIG is drawn for the data-generating process (suppressing officer beliefs), revealing graphically that $Y(d_1)$ is a negative control outcome for D. Panel (b) shows how we may without loss represent V by the relevant principal strata; recall that $Y(d_0) = 0$ in this setting.

would be searched regardless of race, as we saw in Example 1. By reasoning about these mechanisms, it is in this sense that the reimagined view of the outcome test aims to identify discrimination.

An obvious difficulty the analyst faces is that $Y(d_1)$ is only observed for the searched motorists, many of whom were likely subject to discrimination. Due to this selection bias, the use of $Y(d_1)$ as a proxy via the hit rate becomes difficult to tame, as was explored thoroughly in Section 3 as an instance of collider bias. Among other reasons discussed below, this feature of the DGP encourages a partial identification strategy for the causal estimand of interest, which I introduce next.

The negative control interpretation also suggests a conventional proximal identification strategy based on a bridge function (Wang et al., 2018; Ghassami et al., 2026). Because $Y(d_1)$ is binary while V may be high-dimensional, the required restriction is particularly demanding and is not imposed here; in Appendix B.1 I give an intuitive reason why this approach is not viable, related to the calibration of officer beliefs. Since reasoning by principal strata reduces V to a eight-dimensional object without loss, framing the identification argument this way allows the analyst to impose restrictions on the relationship between $Y(d_1)$ and V in a more interpretable and defensible manner.

5.1 Partial Identification of the Probability of Necessity

As argued earlier in the article, much of the outcome test literature lacks a causal estimand, limiting the policy relevance of conclusions about the existence or magnitude of discrimination. In policing, this limitation is especially consequential because discrimination ultimately manifests as harms experienced by motorists. This observation parallels the distinction in the literature on causation and the law between the effects of causes and the causes of effects: while scientific analyses commonly ask how an intervention changes outcomes in a population, legal questions frequently concern whether an exposure caused the outcome experienced by a particular individual (Dawid et al., 2014, 2017; Dawid and Musio, 2022; Cuellar, 2022). Here the corresponding question is: for a minority motorist who was searched, would the search have occurred had that same motorist been perceived as White?

A causal estimand encoding this query is the probability of necessity (Pearl, 1999; Tian and Pearl, 2000)

$$\theta \coloneqq \Pr(D(r_0) = d_0 \mid D = d_1, R = r_1), \tag{13}$$

which is the target of the identification strategy I outline. The choice of θ for the empirical analysis was motivated in part by the decomposition derived in Proposition 2, reproduced here

$$\begin{aligned}
&\mathbb{E}[Y \mid D = d_1, R = r_1] - \mathbb{E}[Y \mid D = d_1, R = r_0] \\
&\qquad = \mathbb{E}[Y \mid \text{AS}, R = r_1] - \mathbb{E}[Y \mid \text{AS}, R = r_0] + \\
&\underbrace{\Pr(\text{PF} \mid D = d_1, R = r_1)}_{\theta}\big\{\mathbb{E}[Y \mid \text{PF}, R = r_1] - \mathbb{E}[Y \mid \text{AS}, R = r_1]\big\} - \\
&\qquad \Pr(\text{AW} \mid D = d_1, R = r_0)\big\{\mathbb{E}[Y \mid \text{AW}, R = r_0] - \mathbb{E}[Y \mid \text{AS}, R = r_0]\big\}.
\end{aligned} \tag{14}$$

Here, θ is the proportion of searched minority motorists whose vehicles would not have been searched had they been perceived as White. This quantity activates the term in (14) describing what I called the productivity of racial profiling. The identification strategy will

impose weak, readily relaxable restrictions on how the proxy $Y(d_1)$ relates to the principal strata, thereby restoring the usefulness of hit rate disparities for indicating the magnitude and direction of discrimination.

My approach will rely on two assumptions, however, below these will be relaxed in an accompanying sensitivity analysis. Let me begin by stating the first assumption.

Assumption 1 (No anti-White bias) $D(r_1) \geq D(r_0)$ *almost surely.*

Assumption 1 is a monotonicity restriction on the effect of race on the decision to search. Few empirical studies have documented evidence suggesting a violation of Assumption 1.[12] Knox et al. (2020) emphasize that the restriction is an approximation: violations could arise if a subset of officers discriminate against White civilians, or if White civilians attract greater suspicion because they appear "out of place" in a predominantly minority neighborhood (Carroll and Gonzalez, 2014; Gelman et al., 2007). Nevertheless, they argue that such cases are plausibly exceptional relative to anti-minority discrimination and show that their substantive conclusions are unchanged after excluding drug-related encounters, which they identify as a particularly plausible setting for violations. I adopt the same substantive interpretation here.

The second assumption controls a narrow aspect of the relationship between the proxy $Y(d_1)$ and the environment $V = (D(r_0), D(r_1))$.

Assumption 2 (No pure inframarginality)

$$\Pr(Y(d_1) = 1 \mid \text{AS}, R = r_1) = \Pr(Y(d_1) = 1 \mid \text{AS}, R = r_0). \tag{15}$$

Assumption 2 concerns only searches that would occur regardless of race. In the stylized example, fragrant searches represent the comparatively productive members of this stratum.

[12]Some studies have claimed to detect anti-White bias for policing outcomes including traffic stops, searches, and police use of force (Antonovics and Knight, 2009; Sadler et al., 2012; Novak and Chamlin, 2012; James et al., 2013; Rosenfeld et al., 2012; Pierson et al., 2020).

More generally, the always-searched stratum may contain both comparatively productive fragrant searches and comparatively unproductive inventory searches. The assumption requires the resulting average hit rate within this stratum to be equal across racial groups. It does not require White and minority motorists to possess contraband at equal rates in the population, nor does it require equal contraband rates among all searched motorists, because the minority searched population may additionally contain profiled furtive searches.

This restriction corresponds closely to an argument made by Ayres (2002). For infra-marginality alone to generate a racial disparity in hit rates without discriminatory selection, White and minority motorists who would be searched regardless of race must differ systematically in their probability of possessing evidence of illegality. Such a pattern could arise, for example, if minority motorists were significantly more likely to have their vehicles impounded for administrative reasons, such as an expired registration or suspended license, while White motorists were more likely to enter the always-searched stratum through a fragrant search. Absent a mechanism producing such a difference, Ayres argues that there are not compelling reasons to expect it.

There is also empirical evidence consistent with the restriction. Tillyer and Klahm (2011), studying a large Midwestern municipal police department, distinguish between mandatory searches, in which department policy leaves the officer little or no discretion over whether a search occurs, and discretionary searches. They find nearly identical contraband recovery rates for White and Black motorists among mandatory searches, but substantial differences among discretionary searches. Similarly, Chanin et al. (2018), studying matched Black and White motorists in San Diego, find almost identical hit rates for searches incident to arrest and no statistically significant difference in inventory-search hit rates. These observed categories need not coincide exactly with the always-searched principal stratum, but they provide evidence in favor of (15).

Complementary evidence is provided by Feigenberg and Miller (2022), who study Texas Highway Patrol searches and find that substantial racial disparities in search intensity are

not accompanied by corresponding improvements in search productivity. Black and Hispanic motorists are searched considerably more often than White motorists, yet their searches are no more likely to recover contraband. Furthermore, variation in troopers' search propensities is largely unrelated to their contraband recovery rates, leading the authors to conclude that officers have limited ability to sort motorists according to search productivity. This result does not directly test Assumption 2, but it provides little indication that race marks large differences in contraband incidence among motorists sufficiently suspicious to be searched regardless of race, weakening one potential basis for violations of the assumption.

With these assumptions in hand, I have the following proposition.

Proposition 5 (Lower bound on the probability of necessity) *For measured covariate X, let $\theta(x)$, $x \in supp(X)$, denote the probability of necessity conditional on x for each $x \in supp(X)$. Under Assumptions 1 and 2 (conditional on x), it follows that $\theta(x) \in [\underline{\theta}(x), 1]$ where*

$$\underline{\theta}(x) = 1 - \min_{y \in \{0,1\}} \frac{\Pr(Y = y \mid D = d_1, R = r_1, X = x)}{\Pr(Y = y \mid D = d_1, R = r_0, X = x)}, \tag{16}$$

is a sharp lower bound on $\theta(x)$.

Proof. See Appendix A.4. ■

The origin of this bound derives from the intuition given above. The observed minority hit rate combines motorists in the always-searched (fragrant) and profiled (furtive) strata. We are interested in how the proxy is informative of the latter stratum. To bound the proportion belonging to furtive searches, we leverage Assumption 2 to control the hit rate among fragrant searches of minority motorists with observed data. It does this by equating the fragrant minority hit rate with that of White motorists, and since under monotonicity every search of a White motorist is fragrant, the hit rate among fragrant White (and thus minority) motorists is identified. In this way, a lower minority hit rate implies that some minority searches must be furtive searches, i.e. attributable to their race, reiterating the proxy intuition outlined in Example 1. If these searches are less productive, their addition

dilutes the minority hit rate. The lower bound imposes the smallest proportion of furtive searches capable of producing the observed disparity, without requiring their hit rate to be known. Notice that this reasoning, in contrast to inferences based on marginal motorists, both deduces the presence of discrimination and quantifies its magnitude.

Operationally, the evidence may appear in either outcome category: contraband may be found at a lower rate among searched minority motorists, or it may be absent at a lower rate among searched White motorists. Taking the smaller of the two likelihood ratios in (16) produces the lower bound (16). The upper bound on θ equals unity because the assumptions we impose do not prevent every observed minority search being racially profiled.

The bound therefore restores the common intuition underlying the outcome test by making a hit rate disparity directly informative about discrimination. It partially identifies a policy-relevant causal estimand in a readily communicable form. Equation (16) expands on a non-sharp bound derived in Appendix A.7 of Knox et al. (2020). The causal model developed here supplies a new interpretation of the outcome test that motivates the identification strategy, while combining the racial differences in hit and miss rates makes the bound sharp.[13] I extend their result further with a novel sensitivity analysis, which is given below.

Concerning estimation of the bounds in (16), the appearance of the non-regular min function introduces some complications. The natural plug-in estimator can exhibit finite-sample bias when the covariate cells are sparse. Suppose for example that the hit rates by race coincide within a given x. Then, the population lower bound $\underline{\theta}(x)$ equals zero. Nevertheless, sampling variation will generally cause one of the two estimated likelihood ratios in (16) to fall below one, inflating the estimated lower bound away from zero, even if the estimators for the numerator and denominator are unbiased. Standard debiasing techniques, such as double machine learning, are not straightforwardly applicable as the minimum is not a pathwise differentiable function due to the vertex at equal hit rates. Empirically, the severity of this

[13]If the minority hit rate is lower than the White hit rate, the bound in Knox et al. (2020) is sharp and equals (16), which will be the case for every Black and Hispanic comparison I report in the empirical application. For Native American motorists, whose hit rate will exceed the White rate, their bound is zero (revealing the lack of sharpness), while (16) is positive.

depends on the resolution of X. Fortunately, ignorability was not a required condition for identification, meaning that conditioning on a fine X is not necessary. The analyst may wish to condition on certain covariates to understand how $\underline{\theta}$ varies by context, however, and examples of which will be given below.

I will proceed by estimating the empirical conditional mean of Y on (R, X) among the searched, where inference will rely on the fact that the minimum is a directionally differentiable function, meaning that under certain conditions familiar limiting distributions for the estimator become available, as shown by Fang and Santos (2019). Based on their work, a fast adaptation of the bootstrap to non-regular functions becomes accessible; I provide implementation details in Appendix C.

Sensitivity analysis Even though there is much empirical merit to support Assumptions 1 and 2, it is plausible that they may not hold absolutely. I next develop a sensitivity analysis which parametrizes the lower bound by the fraction of searches which fail to satisfy each of the stated assumptions above. Let me introduce some notation for conciseness. For $y \in \{0, 1\}$, write the hit rate for race r as $h_r(y \mid x) = \Pr(Y = y \mid D = d_1, R = r, X = x)$, and let $a_r(x) = \Pr\{Y(d_1) = 1 \mid \text{AS}, R = r, X = x\}$ be the success rate of always-search stops. Denote by η the maximum fraction of searches of White drivers which would not have occurred were the driver perceived a minority, that is $\Pr(\text{AW} \mid D = d_1, R = r_0, X = x) \leq \eta$. When Assumption 1 holds, $\eta = 0$. Next, denote by ρ the largest proportional reduction in contraband discovery permitted among always-searched minority motorists relative to always-searched White motorists, such that $(1 - \rho)a_0(x) \leq a_1(x) \leq a_0(x)$. When Assumption 2 holds, $\rho = 0$. For example, $\rho = 0.2$ allows the contraband rate among always-searched minority motorists to be as much as twenty percent lower than that among always-searched White motorists. I have the following proposition.

Proposition 6 (Sensitivity of θ) *Fix $\eta, \rho \in [0, 1)$. Suppose, for every $x \in \text{supp}(X)$, that $\Pr(\text{AW} \mid D = d_1, R = r_0, X = x) \leq \eta$ and $(1 - \rho)a_0(x) \leq a_1(x) \leq a_0(x)$. Then $\theta(x) \in$*

$[\underline{\theta}_{\eta,\rho}(x), 1]$, *where*

$$\underline{\theta}_{\eta,\rho}(x) = 1 - \min\left\{1,\ \frac{1-\eta}{1-\rho}\frac{h_{r_1}(1 \mid x)}{h_{r_0}(1 \mid x) - \eta}, (1-\eta)\frac{h_{r_1}(0 \mid x)}{h_{r_0}(0 \mid x) - \eta}\right\}, \tag{17}$$

is a sharp lower bound for $\theta(x)$*. A ratio in* (17) *is understood to equal* $+\infty$ *when its denominator is nonpositive.*

Proof. See Appendix A.5. ■

Similar to Proposition 5, the minimum is taken by comparing the now adjusted disparity in hit rates (second branch) and miss rates (third branch). To build intuition for (17), it is instructive to relax the assumptions one at a time. First assume $\rho = 0$, so that always-searched motorists are equally productive across racial groups, and consider violations of monotonicity. At $\eta = 0$, the bound matches Proposition (5), and so the same intuition applies. When $\eta > 0$, the searched White population may now contain both always-searched and anti-White searches. Consider a commuter searched for appearing "out of place" in a minority neighborhood. This vehicle produces no contraband and lowers the observed White hit rate, making the minority shortfall harder to explain without racial profiling. Suppose the White driver were instead a courier, since, by contrast, this vehicle produces a hit and raises the observed White rate above the productivity of the motorists who would be searched regardless of race. The bound takes the case most adversarial to evidence of racial profiling: every anti-White search in violation of Assumption 1 is assigned to a courier. Removing these productive searches from the White comparison population lowers the inferred hit rate among the remaining always-searched motorists, allowing some of the observed disparity to be attributed to the composition of White searches rather than to profiled minority searches.

How much this alternative explanation matters depends on the observed White hit rate. When $\rho = 0$, let q_1^η denote the second branch of the minimum in (17), suppressing x for

simplicity. Its sensitivity to η is given by

$$\frac{\partial \log q_1^\eta}{\partial \eta} = \frac{1}{h_{r_0}(1) - \eta} - \frac{1}{1 - \eta}, \qquad \left.\frac{\partial \log q_1^\eta}{\partial \eta}\right|_{\eta=0} = \frac{1 - h_{r_0}(1)}{h_{r_0}(1)}. \tag{18}$$

The expression on the right is the odds of a miss to a hit among searched White motorists. When the White hit rate is high, these odds are small. Even in the DGP where every anti-White search is of a courier, many hits remain among White motorists who would be searched regardless of race. Their inferred hit rate therefore remains relatively high, leaving much of its disparity with the minority hit rate unexplained by a violation of monotonicity. I highlight (18) because it is primes expectations on how robust an empirical analysis will be when $\eta > 0$. The lower bound is consequently least sensitive to violations of monotonicity where White searches are already highly productive, and most sensitive where they are not. I present a numerical illustration of this intuition in the empirical application.

Next maintain $\eta = 0$ and relax Assumption 2. Here, consider the contrast between fragrant and inventory searches. Fragrant searches are comparatively more likely to recover contraband. If always-searched White motorists disproportionately face fragrant searches while always-searched minority motorists disproportionately undergo inventory searches, minority motorists may have a lower hit rate even among searches that were not caused by race.

The parameter ρ permits the always-searched minority hit rate to be as much as a proportion ρ lower than its White counterpart. I consider this direction because it is the one capable of explaining the empirically relevant shortfall in the minority hit rate. Allowing always-searched minority motorists to be more productive than their White counterparts would instead make that shortfall harder to reconcile with the less racial profiling: more of such searches raises the minority hit rate, so an even larger share of unproductive furtive searches would be required to pull the observed rate downward. The observed shortfall arises then either because the always-searched minority population contains a less productive set of

non-discriminatory searches, or because the searched minority population additionally contains searches attributable to race. Increasing ρ widens the scope for the former explanation.

Should it be the case that for some minority group—an example of which will be provided the application—the hit rates are consistently higher than those of White drivers, it may suit the analyst to relax Assumption 2 in the other direction. To do so, let $\kappa \geq 1$ be such that

$$\Pr(Y(d_1) = 1 \mid \text{AS}, R = r_0) \;\leq\; \Pr(Y(d_1) = 1 \mid \text{AS}, R = r_1) \;\leq\; \kappa \Pr(Y(d_1) = 1 \mid \text{AS}, R = r_0), \tag{19}$$

so that always-searched minority motorists may recover contraband up to $100(\kappa - 1)\%$ more often than the always-searched White rate. Each reproduces Assumption 2, at $\kappa = 1$ and at $\rho = 0$ respectively. Accounting for this relaxation, I prove an alternative of Proposition 6 in Appendix A.5.

6 Empirical Application

For an empirical illustration, I apply the results of the previous section to recorded stops and searches conducted by the Washington State Patrol between 2009 and 2018. These data form part of the Stanford Open Policing Project and have been analysed by Pierson et al. (2020), who apply the veil-of-darkness test to stop decisions and the threshold test of Simoiu et al. (2017) to search decisions. As discussed in earlier sections, the latter test fails to quantify the harm inflicted by discriminatory policing in an interpretable manner. In this section, I build on their results to remedy this problem, including through an analysis of whether a major policy change, the legalization of marijuana, affected the extent of discriminatory policing.

6.1 Data

The data contain 11,333,425 vehicle stop records from 2009–2018, all vehicular stops conducted by a single state agency. Features of the stop include date, time, county, subject race, sex and age, the violation charged, and arrest, citation, warning, search, search basis, and contraband recovery indicators. I code race as White, Black, Hispanic, Native American or Other, the last pooling the Asian and Pacific Islander and residual categories recorded by WSP. Contraband recovery is recorded without any missingness for every searched motorist.

Table 4: Stops, search rates, and hit rates by race and legalization era

Era	Race	# Stops	Search rate (%)			Hit rate (%)		
			All	Low Disc.	High Disc.	All	Low Disc.	High Disc.
2009–2012	Black	147,100	6.70	6.35	0.35	16.8	14.4	60.3
(Pre-period)	Hispanic	295,785	5.16	4.83	0.33	13.4	11.1	45.9
	White	2,641,131	3.03	2.85	0.19	19.0	15.8	67.7
	Native	23,252	13.17	12.33	0.84	18.8	15.2	72.3
	Other	218,509	2.53	2.42	0.12	11.7	10.1	45.8
2013–2018	Black	246,581	4.49	4.36	0.13	7.3	5.9	53.5
(Post-period)	Hispanic	479,275	3.79	3.63	0.16	6.9	5.2	45.2
	White	3,480,365	2.32	2.24	0.09	8.8	6.7	63.9
	Native	30,740	10.66	10.25	0.41	11.1	8.7	73.0
	Other	365,710	1.85	1.80	0.05	5.4	4.0	51.3

The legal basis for a search is recorded as a single-letter code. The standardized field distributed by the Open Policing Project collapses three of these into a residual category absorbing 95% of searches, so I work from the raw code. These are: incident-to-arrest (219,745 searches, 9.6% hit rate), impound (7,571, 8.4%), warrant (2,098, 91.8%), consent (11,089, 62.5%), and canine (703, 56.6%). I group the first three as low-discretionary and the last two as high-discretionary. Washington does not recognize the automobile exception to the warrant requirement, so an officer with well-founded probable cause still cannot search a vehicle without a warrant and must instead rely on another exception, in practice the driver's consent (State v. Snapp, 2012). Discretion over whether to search therefore operates almost entirely through consent searches.

Race is unrecorded for 3,193,337 stops (28.2%) and no proxy information such as the subject's surname is available to mitigate this (McCartan et al., 2025; Cooper et al., 2026b); these records were removed. Following Pierson et al. (2020) I drop the fourth quarter of 2012 (211,640 stops), which straddles the legalization date of 6 December 2012. After these cleaning decisions, 7,928,448 stops, or 70.0% of the original data, remained. County is unrecorded for a further 1,084,339 of these, which are excluded only from the analysis by side of the Cascade crest.

Major covariates comprise the legalization era (pre- and post-marijuana policy change in December 2012) and geography. Geography is coded east or west of the Cascade crest, which separates the urban western counties containing Seattle, Tacoma and Everett from the rural and agricultural east. I do not consider the reason for the stop, as it is unrecorded for much of the data, a limitation Pierson et al. (2020) also note. In Table 4 I give some important summary statistics by era. Note that hit rates in low-discretionary searches are close for Black and White motorists in both eras, providing support for Assumption 2. Furthermore, minority motorists are consistently searched more frequently than White motorists, meaning that we do not falsify Assumption 1 (no anti-White bias).

Native American motorists are searched at 13.2% of stops before legalization and 10.7% after, around four and a half times the White rate in both eras. Their hit rates are close to the White rate in the pre-period and higher in the post-period, and in high-discretionary searches they are the highest of any group, at 72.3% before legalization and 73.0% after. This combination is consistently found by year and has been widely reported (Criscione and Henshaw, 2025).[14] There, interpreting these search disparities to be too large to be justifiable, analysts have sought to find evidence of discrimination using conventional approaches to the test, where a lower hit rate would be expected as evidence of discrimination—as reinforced by recent versions of the test (Gaebler and Goel, 2025). This article has argued that this conventional intuition for the test should be adopted cautiously, because it relies on strong

[14]Their analysis sample concerns 2018-2023, however the empirical pattern I find here is identical.

assumptions. Below, the framework of this paper will provide evidence that this empirical pattern is consistent with discrimination against Native Americans.

6.2 Results

Restricting to high-discretionary searches, I find a lower bound on the probability of necessity θ of 13.0% (95% confidence limit: 8.6%) for Black motorists, 31.2% (28.2%) for Hispanic motorists and 18.7% (6.2%) for Native American motorists over the entire period, indicating that at least one in eight searched Black motorists, close to one in three searched Hispanic motorists and one in six searched Native motorists would not have been searched were it not for their race.

Washington's Hispanic population is concentrated in the agricultural counties east of the Cascade crest, while its Black population is concentrated in the western metropolitan counties. Theories of "out-of-place" policing hold that officers use race as a cue for incongruence with the perceived social character of a neighborhood, generating heightened scrutiny when an individual appears not to suit local expectations (Meehan and Ponder, 2002; Supreme Court of Arizona, 1975; Carroll and Gonzalez, 2014; Novak and Chamlin, 2012). The natural discontinuity in demography across the Cascades allows us to test for evidence that these mechanisms of discriminatory policing operate. On the other hand, an alternative mechanism for producing racial disparities in stops, referred to as "in-place" policing, whereby officers conduct more stops within minority neighborhoods because such areas are treated as higher-crime environments and are therefore targeted for enforcement aimed at detecting contraband, where often low-level offenses such as speeding are used as a 'pretext' to search a civilian (Gelman et al., 2007; Petrocelli et al., 2003; Epp et al., 2014; Pierson et al., 2020). For Hispanic motorists the bound is 46.1% west of the crest (confidence limit 41.7%) against 26.3% (21.7%) east of it, strongly indicative of the former account, while for Black motorists (13.6% west against 14.7% east) and Native motorists (19.4% against 15.4%) the intervals overlap such that there is no strong evidence for either mechanism.

Table 5: Lower bounds on θ among high-discretionary searches, by race and legalization era. White is the reference group, so no bound is defined for it.

	Pre-legalization			Post-legalization		
Race	N	Hit rate (%)	$\hat{\underline{\theta}}$ (95% CI)	N	Hit rate (%)	$\hat{\underline{\theta}}$ (95% CI)
Black	516	60.3	0.110 (0.056)	318	53.5	0.164 (0.094)
Hispanic	988	45.9	0.323 (0.283)	765	45.2	0.293 (0.245)
Native	195	72.3	0.142 (0.000)	126	73.0	0.252 (0.076)
White	4,931	67.7	—	2,989	63.9	—

Marijuana legalization Legalization of marijuana removed a common predicate for searching a vehicle, and the share of stops resulting in a discretionary search at least halved for every racial group, from 0.35% to 0.13% for Black motorists, 0.33% to 0.16% for Hispanic motorists, 0.84% to 0.41% for Native motorists and 0.19% to 0.09% for White motorists (see Table 4). However, despite this, we do not find reason to be optimistic about a decrease in racial profiling. In Table 5, I report the bound by era: 11.0% before legalization and 16.4% after for Black motorists, 32.3% and 29.3% for Hispanic motorists, and 14.2% and 25.2% for Native motorists. Although the estimates are intervals, the absence of any decline in their lower endpoints indicates that the minimum amount of discrimination consistent with the data did not fall between eras, which explains why this evidence leaves us pessimistic about an improvement in officer behavior.

Sensitivity analysis Next, I assess the robustness of the above conclusions to violations of Assumptions 1 and 2. Figures 6 and 7 plot $\hat{\underline{\theta}}_{\eta,\rho}$ and its sensitivity uncertainty as each parameter varies in turn. Table 6 records the largest violation of each compatible with a bound that remains distinguishable from zero.

For Hispanic motorists the confidence interval for the lower bound excludes zero until $\eta = 0.36$, which would require that more than one in three searched White motorists were searched only because they were perceived as White, or until $\rho = 0.29$, requiring always-searched Hispanic motorists to recover contraband at roughly seven-tenths the White rate.

For Black motorists, results are robust until $\eta = 0.15$, or until $\rho = 0.09$. Native results are considerably less robust to monotonicity than the Black ones, with the confidence limit excluding zero only until $\eta = 0.04$. For Native motorists, however, hit rates tend to exceed those of White drivers, meaning it is sensible to relax Assumption 2 in the opposite sense to that used for Black and Hispanic drivers, via (19), and Table 6 reports κ in addition to ρ. The lower confidence limit is already at zero before legalization and remains above zero only until $\kappa = 1.05$ after. Over the full period it reaches zero at $\kappa = 1.04$.

These results describe robustness to the presence of discrimination, Table 6 also reports when I am no longer able to conclude a moderate amount of discriminatory behavior is present, which I set at a rate of 10% of searches. The Hispanic results remain strongly robust where we cross that threshold at $\eta = 0.28$ and $\rho = 0.20$, while the Black and Native confidence limits lie below 10% already at $\eta = \rho = 0$, so no conclusion of moderate discrimination is available for either group. The Hispanic results remain robust within legalization era and geographic region, and to joint relaxation of the two assumptions, see Table 7 in Appendix D for numerical results of the latter.

Nearly all high-discretionary searches in Washington proceed by consent, so a reader may be concerned that differential rates of refusal by race affect my estimates. Consent refusal alone does not affect the bound because for the searches we analyse, a search can occur if and only if consent was granted. A violation of Assumption 2 arises only if refusal is correlated with guilt differentially by race. If minority motorists carrying contraband decline more readily, relative to those who are not, than White motorists do, then this would weaken evidence of discrimination. A large disparity of that kind is unlikely because the incentive to decline is dominated by the prospect of arrest, which is common to all motorists. Nevertheless, ρ accounts for some of this behavior: selection of this kind leaves always-searched minority motorists recovering contraband at a lower rate for reasons unconnected to the officer, and Table 6 shows that it would need to be pronounced before the Hispanic conclusions are invalidated, though the Black results are overturned by a reduction of less

Table 6: Largest violations of each assumption required to explain away evidence of discrimination. The columns under CI $> c$ state the largest values of η, ρ and κ that are required before the confidence limit on $\underline{\theta}$ includes c. Assumption 2 may be relaxed in either direction, either via ρ or κ. In this table κ is not relaxed for Black and Hispanic searches as it is not the direction of relaxation consistent with empirical observation. See the Appendix for additional relaxation of their results via κ. An entry of — indicates that the limit remains above the level throughout $[0, 1)$ for η and ρ, or throughout $[1, 2)$ for κ. An entry of 0.00 indicates that the limit lies below the level already at $\eta = \rho = 0$.

					CI > 0			CI > 0.10		
	Race	N	$\hat{\underline{\theta}}$	95% CI	η	ρ	κ	η	ρ	κ
All years	Black	834	0.130	0.086	0.15	0.09	×	0.00	0.00	×
Pre-legalization	Black	516	0.110	0.056	0.11	0.06	×	0.00	0.00	×
Post-legalization	Black	318	0.164	0.094	0.14	0.10	×	0.00	0.00	×
East of Cascades	Black	211	0.147	0.064	0.12	0.07	×	0.00	0.00	×
West of Cascades	Black	487	0.136	0.080	0.13	0.08	×	0.00	0.00	×
All years	Hispanic	1,753	0.312	0.282	0.36	0.29	×	0.28	0.20	×
Pre-legalization	Hispanic	988	0.323	0.283	0.37	0.29	×	0.30	0.20	×
Post-legalization	Hispanic	765	0.293	0.245	0.30	0.25	×	0.22	0.16	×
East of Cascades	Hispanic	825	0.263	0.217	0.31	0.22	×	0.21	0.13	×
West of Cascades	Hispanic	695	0.461	0.417	0.44	0.42	×	0.39	0.35	×
All years	Native	321	0.187	0.062	0.04	×	1.04	0.00	×	1.00
Pre-legalization	Native	195	0.142	0.000	0.00	×	1.00	0.00	×	1.00
Post-legalization	Native	126	0.252	0.076	0.05	×	1.05	0.00	×	1.00
East of Cascades	Native	148	0.154	0.000	0.00	×	1.00	0.00	×	1.00
West of Cascades	Native	106	0.194	0.000	0.00	×	1.00	0.00	×	1.00

than a tenth.

Monotonicity is a strong assumption and is a powerful tool for tractable identification of a target quantity. For that reason, it may seem surprising that the results above are insensitive to non-trivial violations of the assumption. In Section 5.1, I gave some intuition for this behavior, which showed that when officers are much more successful than not when searching White motorists, the bound is robust to changes in η when η is small. Using data taken directly from the analysis, I illustrate this intuition with an example. Over the full period, 1,753 Hispanic motorists were searched and 799 were found carrying contraband, against 7,920 White motorists searched and 5,250 carrying. These figures produce hit rates of 45.6% and 66.3% respectively, applying (16) yields $\underline{\theta} = 0.312$. Setting $\eta = 0.05$ concedes

that 396 of those 7,920 White searches occurred only because the driver was perceived as White, and, taking the case most adversarial to a finding of profiling, that every one of the 396 recovered contraband. Removing them leaves 4,854 recoveries among 7,524 always-searched White motorists, a rate of 64.5% rather than 66.3%, and the bound falls only to 0.294; at $\eta = 0.10$ it falls to 0.271. Two in three White searches were productive to begin with, so withdrawing a handful of productive searches from largely successful searches of White drivers minimally affects the hit rate. Eliminating the bound would require 3,014 of the 7,920 White searches to have been anti-White, every one of them productive. Of course, it is unlikely for this to occur, explaining why the results are robust to these hypothetical violations of monotonicity.

I will now briefly discuss how to interpret the magnitude of these sensitivity parameters. Both parameters control unobservable quantities and are thus difficult to tie directly to observed data, however, there are arguments that can be made which guide how large or small they should be considered to be. Anti-White searches are generally held to be rare, so η is expected to be near zero. The exercise above is therefore not an argument that any particular η is credible; it measures how far the conclusions rest on an assumption that is mathematically strong but empirically uncontroversial. For reasons discussed above, my results do not depend on this assumption strongly.

Assumption 2 however, can absorb reasons even unconnected to the officer for why motorists are carrying contraband at different rates. One strategy to assess these reasons might be to consider how consumption differs by group. National surveys on self-reported marijuana usage suggest that Black and White people consume at similar rates, while Native Americans do so noticeably more (Substance Abuse and Mental Health Services Administration, 2024). This could motivate requiring robustness to a large κ, implying that the results above for Native Americans are highly sensitive to Assumption 2.

However, these data should be interpreted with caution as they pertain to the parameters ρ and κ. Recall that the unit of analysis is the police stop. First, then, it is important to

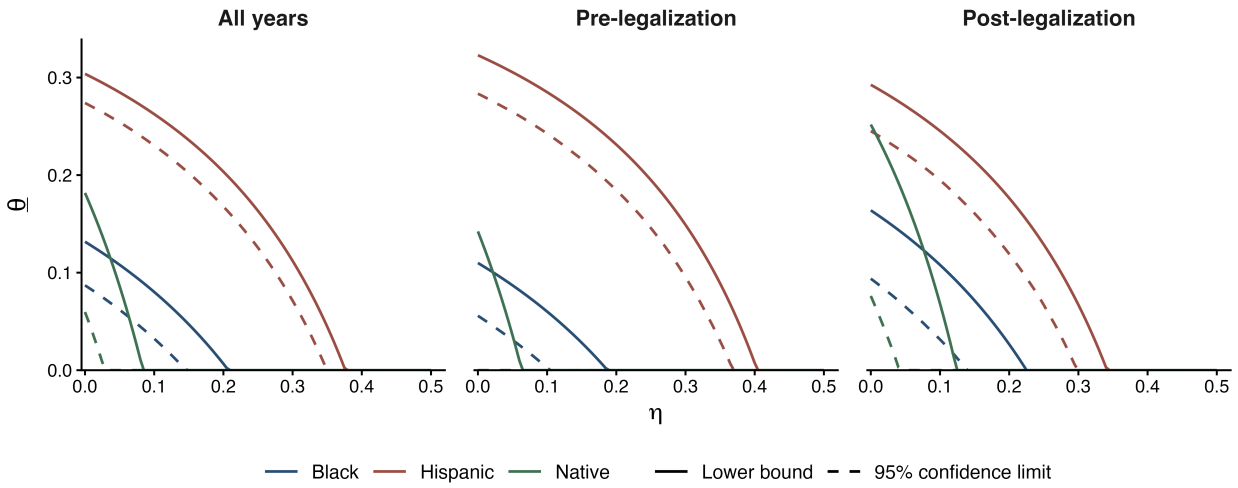


Figure 6: **Sensitivity of the lower bound to violations of Assumption 1.** Solid lines give $\underline{\hat{\theta}}_{\eta,0}$ and dashed lines the one-sided 95% confidence limit, as η varies with ρ held at zero.

note that consuming more is not the same as transporting more unlawfully. Marijuana possession or consumption was never unrestrictedly lawful *in a vehicle* (e.g. to prohibit impaired driving), and that additional restriction on behavior may close much of the gap the consumption figures imply. Second, if marijuana usage rates were driving the finding, then since the policy change in 2012 removed most of it from the definition of contraband, the bound should have fallen; instead it rose across the two eras, 14.2% before against 25.2% after, and the result is in fact harder to overturn afterwards ($\kappa = 1.05$ against 1.00). Third, the restriction is stated among always-searched motorists, relative to comparable White drivers. These kinds of searches are conventionally regarded as officers requiring suspicion well in excess of the typical "reasonable suspicion" standard that is applied to search a vehicle. Overturning the result requires those Native motorists to recover contraband between three and five per cent more often than comparable White motorists, which is difficult to motivate. Nevertheless, carrying rates by race are not recorded in these data and cannot be recovered from them, so these arguments serve to guide discussion of plausibility; designing approaches to recover direct evidence of guilt is a question for future work.

Comparison with the threshold test Pierson et al. apply the threshold test to these

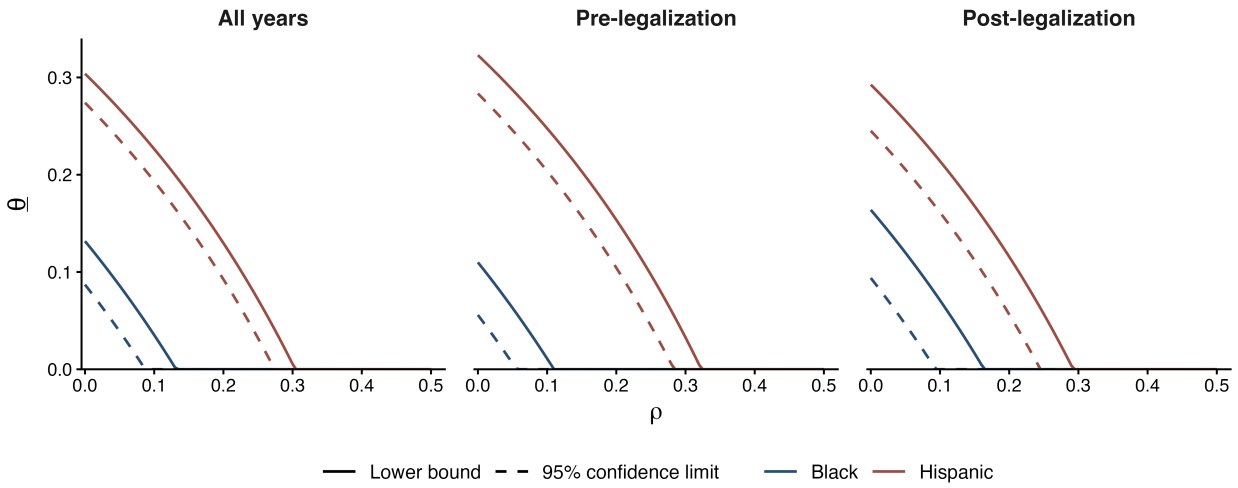


Figure 7: **Sensitivity of the lower bound to violations of Assumption 2.** Solid lines give $\underline{\hat{\theta}}_{0,\rho}$ and dashed lines the one-sided 95% confidence limit, as ρ varies with η held at zero. Native motorists are omitted because ρ is interpreted on a different scale to κ; the corresponding plots as κ varies are given in the Appendix.

data and find that White drivers faced consistently higher inferred search thresholds than Black and Hispanic drivers, both before and after legalization. My estimates agree in direction but build on their results by giving an interpretable measure of their strength.

The analysis of officer behavior before and after the police change reveals the practical value of θ most clearly. Pierson et al. find their inferred thresholds fell for all racial groups after legalization, and offer three candidate explanations: that officers had not fully internalized the policy change and continued searching drivers who were no longer high risk; that they redirected attention toward more serious offences and applied a threshold commensurate with their severity; or that officers relieved of policing marijuana possession had more resources and spent them on lower yield searches. None of these mechanisms are identifiable in the data, however, the analysis presents a finding which is consistent with each, directly tied to the harms felt by civilians in the wake of the policy change.

Indeed, legalization removed a category of contraband that officers could search for, and predictably, search rates decreased after the policy change. Moreover, searches became less successful and the effect is visible in the hit rates, which fall for every group after 2012. That said, the ratio of minority to White hit rates moved little, standing at 0.89 and 0.68

before legalization and 0.84 and 0.71 after, for Black and Hispanic motorists respectively. Altogether, these data suggest that officers did not materially use race as a proxy for the possession of marijuana, weighing against explanations of the results that appeal to statistical discrimination. Rather, the general lack of a decrease in $\underline{\theta}$ implies that even if officer behavior predictably changed due to the policy, racial profiling remained embedded in the practices employed by officers in encounters with civilians. This is of course except for Native American motorists, for whom the bound noticeably increased.

7 Discussion

7.1 Related Work

Two recent theoretical contributions are close to the framework developed here. In this subsection, I will discuss similarities with their work and distinguish my contribution. Hull (2021), hereafter HWP,[15] and, Canay et al. (2024), hereafter CMM, study what marginal outcome tests can reveal when the analyst leaves the decision-making model largely unspecified. This is in contrast to a trend in economics to specify the decision-maker as solving some kind of optimization problem (Durlauf, 2006). Both papers ultimately arrive at versions of the threshold representation that underpins much of the outcome-test literature, as discussed in Section 2.

I begin with general comments. The framework I present here subsumes both of these models, and, by doing so does not take them as primitive. I begin instead from the causal model (2), placing no restrictions on the environment V, and a minimal restriction on the formation of beliefs B. Beliefs interpreted as predictions of the outcome are shown to arise with minimal loss of generality within this model, and threshold models used throughout the outcome-test literature are subsequently derived from it. This ordering is useful because assumptions concerning belief formation (and its corresponding state space), belief accuracy,

[15]HWP abbreviates Hull Working Paper.

and the information permitted to affect the decision can be stated separately, rather than being built into a particular decision rule from the outset.

Further, I give causal definitions to the principal forms of discrimination discussed in this literature, with taste-based and statistical discrimination corresponding to mediation effects. These mechanisms may operate simultaneously, and neither definition requires beliefs to be calibrated. Disparities in officer thresholds remain contained within the model and may be targeted when substantively interesting, but I discourage their use to define discrimination. This permits the analysis to move beyond asking whether racial groups face different decision thresholds toward causal questions concerning whether race changes decisions, through which mechanisms, and by how much.

The scope of my analysis also broadens that of HWP and CMM. There, the authors focus principally on marginal outcome tests, asking what the outcomes of individuals at the decision margin reveal about discrimination. I additionally dissect the traditional hit rate version of the test and its connection to the marginal motorist. I leverage simple and familiar causal concepts such as principal stratification, causal interaction, and collider bias to provide novel and intuitive explanations for why hit rates are generally difficult to interpret, and clarify the restrictions under which they become informative. The results throughout this article consequently provide a set of warnings for empirical deployments of both versions of the outcome test. Several of these warnings are tacitly acknowledged by the scholarly community but largely ignored in existing empirical work; one major contribution of this article is to collect them within a common causal model and clearly identify where each enters the logic of the test.

Next, I discuss some technical differences between our work. HWP assumes that there exists a continuous latent ranking U such that $D = \mathbf{1}\{U \leq 0\}$, and defines the marginal outcome test by

$$\beta_{\text{HWP}} = \mathbb{E}[Y^* \mid U = 0, R = r_0] - \mathbb{E}[Y^* \mid U = 0, R = r_1]. \tag{20}$$

His central question is how this contrast should be interpreted. HWP defines accurate statistical discrimination by the existence of some signal $S \subseteq V$ such that $U = \mathbb{E}[Y^* \mid R, S] - t$ for a common threshold t, and canonical taste-based discrimination by replacing t with $t(R)$. Behavior admitting neither representation is classified as biased beliefs. These definitions permit HWP to characterize what marginal outcomes—as identified by marginal treatment effects—reveal about the behavioral model generating the ranking.

It is on this point that HWP and I conceptually diverge. HWP examines which behavioral model can rationalize the ranking U. This necessarily creates competing explanations for observed behavior: canonical taste-based discrimination and accurate statistical discrimination are by definition mutually exclusive. I argue in Section 2.1 why this need not hold, even conceptually. To avoid this, I instead define these types of discrimination as mediated effects, which decompose racial bias. Taste-based (NDE) and statistical discrimination (NIE) can therefore present simultaneously, with the calibration of the beliefs involved a separate property. This distinction is consequential for interpreting empirical results: adopting my characterization of discrimination, current empirical work cannot attribute how much of the observed decision-making is attributable to either kind of bias; they are assumed to manifest separately.

The focus of the CMM work is related. They begin by showing that the decision making process can be cast into a Generalized Roy model (GRM) (Heckman and Vytlacil, 2007). There, racial bias in bail decisions grounds the theoretical model. Let $\Delta = Y(d_1) - Y(d_0)$ denote the effect of release on misconduct. The GRM characterizes the decision of judge z as follows, where

$$D_z = \mathbf{1}\{\mathbb{E}[\Delta \mid r, v] \leq \tau(z, r, v)\}, \qquad \tau(z, r, v) = c(z, r, v) + \lambda(z, r, v) + \beta(z, r, v). \tag{21}$$

Here c captures the perceived cost of detention, λ the discrepancy between the objective outcome and the judge's subjective prediction of it, and β a taste for discrimination. Thus,

everything that may alter the judge's benefit (or willingness) to release the defendant is collected in τ, which is compared to their actual likelihood of recidivism.

Their definitions of bias are inherited from this model. CMM distinguish racial β-bias, defined by a nonconstant $\beta(z, r, v)$ in r, from racial τ-bias, defined by a nonconstant $\tau(z, r, v)$ in r. They further distinguish local and global versions of these notions with respect to v; if the racial preferences are found to cross over v, they say that the judge's bias is unclassified. This produces a collection of estimands whose substantive meaning depends heavily on the chosen additive representation. In particular, τ-bias combines racial preferences, prediction and measurement errors, and other considerations entering the decision. While we agree that a difference in β reflects taste based discrimination as defined in this article, differences in τ are hard to interpret as measures of bias and generally do not correspond to a meaningful causal estimand.

The primary pursuit of CMM is to illustrate the logical invalidity of the test when specified as the GRM above. They argue that an Extended Roy Model (ERM), which requires

$$D_z = \mathbf{1}\{\mathbb{E}[\Delta \mid r, v] \leq \tau(z, r)\}, \tag{22}$$

admits a logically valid outcome test. The difference between (21) and (22) is that V is removed from the perceived benefit of release. This seemingly small change is the basis of their principal result. Under the GRM, CMM show that marginal outcome differences can have any sign regardless of whether the judge is unbiased, biased against White defendants, or biased against Black defendants according to their definitions. Under the ERM, by contrast, the marginal outcome difference equals $\tau(z, r_0) - \tau(z, r_1)$, restoring what they call the logical validity of the outcome test.

In my model, the distinction on which this result hinges is simply whether V is a parent of D in the causal model. The ERM prohibits this: once race and beliefs are fixed, the remaining characteristics of the scenario cannot independently affect the decision. CMM give

an example in the context of bail hearings: a violation of this exclusion could consist of a bail judge who is more willing to release an employed parent because detention would disrupt a job and family obligations, even if employment and family status convey no additional information about misconduct. As noted in Section 2, in the setting of police stops and searches, were an an officer instructed by their commander to search red hatchbacks more frequently, they would act upon this instruction in a manner that bypasses their beliefs about the presence of contraband.

The framework presented here imposes this exclusion restriction from the outset. Thus, all contributions of this article are downstream of the point where the primary analysis of CMM ceases. I argued in Section 2 that this particular exclusion restriction can be deduced necessary from the specification of the causal model; many of the inferential delicacies that undergird the outcome test persist in spite of this assumption.

There is a related difference in the treatment of heterogeneity. CMM's MTE analysis and provided intuition proceeds with scalar V, allowing defendants to be ordered relative to special marginal values $v^*_{z,r} \in \text{supp}(V \mid R = r)$, together with shape restrictions that generate a unique indifference point to make for tractable inference. I find this to be a particular unintuitive manner in which to describe the test as V will in reality never admit such a form. Reasoning via principal stratification has the powerful ability to avoid such limitations and make fewer assumptions, while facilitating a simpler description of the weaknesses of the test.

This expositional distinction becomes more pronounced in CMM's treatment of the traditional hit rate version of the outcome test. Their analysis treats the problem primarily through inframarginality. With scalar V, they consider assumptions including $V \perp\!\!\!\perp R$ and $\mathbb{E}[\Delta \mid R, V] = \mathbb{E}[\Delta \mid V]$. Neither suffices separately for identification, but together, alongside their shape restrictions, they render the average outcome comparison informative about global τ-bias. The presentation of these results is similar in spirit to Proposition 3. However, this is a limited restoration of the test. The restrictions serve to strongly eliminate hetero-

geneity by race in V and $Y(d_1)$ conditional on V; indeed, CMM note that once they are imposed the outcome data become redundant, since the benchmark comparison of decision rates has the same sign. Note, however, that this particular conclusion reaffirms my argument that this literature lacks focus on a causal estimand, since these authors reason that the existence of discrimination implied by one identification strategy makes another redundant. This ignores the value one strategy might offer over another in identifying and quantifying a causal estimand, as I show in Section 5. Finally, like the equilibrium restrictions in KPT, the test becomes informative by placing the underlying heterogeneity in a particularly favorable configuration.

The analysis here instead gives a characterization of why hit rates are difficult to interpret without removing that heterogeneity, contextualizing applications of the test in realistic data environments. Furthermore, by re-evaluating the intuition of the outcome test by treating $Y(d_1)$ as a causal proxy, I move beyond the primitive difficulties of the test to provide an easily interpretable use of the hit rate disparity that does not rely on strong assumptions.

Both papers echo parts of the broader set of warnings developed here. HWP emphasizes the observational equivalence of tastes and miscalibrated beliefs, while CMM allow prediction error, measurement error, and omitted decision-maker objectives and show that some of these phenomena can invalidate an outcome test even when unrelated to race. These are important observations. My aim is to place them within a single causal account and, in several cases, give them greater force.

7.2 Future Work

A major contribution of this article is the spotlighting of a causal estimand whose identification strategy restores the core intuition of the hit rate disparity under mild assumptions, which can be straightforwardly relaxed using modern methods of sensitivity analysis. I argued how this intuition derives from the reevaluation of the success of the decision $Y(d_1)$ as a proxy for the qualifying background context leading to the decision. In the example of police

stops and searches, the proxy identification argument is limited by what is called a lack of completeness (Tchetgen Tchetgen et al., 2024): the cardinality of $Y(d_1)$ is insufficient to capture relevant variation in V. To make progress, I used a principal stratification argument to consider an equivalent formulation of V that is 6-dimensional, i.e. $V = (D(r_0), D(r_1), Y(d_1))$, and by controlling the relationship between $Y(d_1)$ and $(D(r_0), D(r_1))$ I derive the bound in Proposition 5. Future work will broaden the application of this reasoning to other estimands in other contexts where more proxies like $Y(d_1)$ are available, so that tighter bounds or even point identification is possible.

Another major contribution of this article is general causal model given in Section 2. There, I introduce of a graphical notion of the beliefs of the decision-maker, allowing for a clean decomposition of discrimination into its taste-based and statistical varieties as defined by Definitions 2 and 3. Future work in this area would study what identification strategies permit analysts to isolate these quantities empirically. There are two key obstacles to this endeavor, however.

The first is that beliefs cannot be measured accurately. While Proposition 1 shows that the causal model admits a representation allowing for a representation of the decision-maker's belief, eliciting this quantity from observed data is challenging, and strategies to do so will vary heavily between empirical applications. Even in an experimental setup, extracting the beliefs of participants is known to be a delicate task, which is highly subject to the stimulus used and environment studied (Schotter and Trevino, 2014; Schlag et al., 2015; Mummolo and Peterson, 2019), making theory building for causal modeling a daunting task. Owing to the decomposition of the hit rate in Proposition 2, it seems true that reasoning about the relationship between officer beliefs about the productivity of racial profiling might prove useful in this pursuit. I offer some additional thoughts on this idea in Appendix B.2 which studies how the bound derived in Proposition 5 varies with the misspecification of officer beliefs.

The second obstacle is similar to that which motivates the outcome test: many relevant

qualifying characteristics of the decision are unknown to the analyst, leading to the presence of unmeasured confounding. It is well-known within the causal literature that mediated effects are challenging to identify in the presence of unmeasured confounding without very strong assumptions (Imai et al., 2010), meaning that even if beliefs could be measured without error, identification would remain a problem.

There are some potential avenues forward. An obvious first solution to the measurement problem is to study a decision-maker whose beliefs can be accurately measured, that is, a machine. The causal model I describe is highly amenable to the study of algorithmic bias, where the features the model uses to make its predictions can be tightly controlled by the researcher. In so doing, concerning questions of human-AI interaction, scholars may study in what sense and to what degree algorithms have a 'taste' for discrimination, and to what extent the relative balance of taste-based and statistical discrimination affects the human decisions which use its output.

In fact, an interesting consequence of my definition of taste-based discrimination follows from the relationship between mediation and principal strata effects (VanderWeele, 2008, Proposition 1). In the absence of taste-based discrimination, that is, when the natural direct effect of R on D vanishes, it follows that for any r,

$$\mathbb{E}[D(r_1) - D(r_0) \mid B(r_1) = B(r_0)] = 0. \tag{23}$$

An intuitive result: if race cannot directly affect decisions, among the subset of units for whom race leaves beliefs unchanged, the effect of race on decisions must vanish. This provides a test for the presence of taste-based discrimination which is robust to miscalibrated beliefs. That is, if a model is built such that its recommendations are based solely on predictions of some outcome Y, this test uncovers if the model is in fact adhering to that principle. The possible uses of this design are widespread, prominent examples range from bail bond recommendations or pretrial parole in criminal justice to treatment recommendations in a

medical setting. One particular interpreted meaning of an algorithm's taste for discrimination is seen in the context of racial bias algorithmic hiring. For instance, when B represents future worker productivity or job performance, rejecting (23) suggests that a model could be implicitly assessing features such as 'cultural fit' to make its recommendation. While there is a wealth of research on algorithmic bias spanning the social sciences, and computer science (Barocas et al., 2023; Plecko and Bareinboim, 2024; Mitchell et al., 2021), the precise causal tether between the concept of taste-based and statistical discrimination in this domain is novel.[16]

Where beliefs cannot be measured, the above suggestions no longer apply. One avenue forward for future work in this pursuit is the indirect measurement of beliefs, via some proxy. Recent work has studied identification in mediation designs when mediators and (possibly confounders) are unobserved (Dukes et al., 2023; Ghassami et al., 2025; Liu et al., 2025), using concepts from the burgeoning literature in proximal causal inference. There, the idea is not dissimilar to those presented in Section 5, in that the existence of sufficiently rich predictors of beliefs, satisfying some exclusion conditions, are useful in recovering the distribution of the hidden variable. I give some possible examples. In policing, to surmise the sentiment of the officer prior to search, an analysis of the visual and audio features of body camera footage would serve as such a proxy. Similar examples include those based on written documents, such as a police report,[17] or in court, this could be the transcript of the proceedings, likewise for a job interview or medical report.

To model the measurement error between the proxy and the belief, large language models (LLMs) can mitigate subjectivity in the review of human language and behavior, while also substantially reducing labor cost of the analysis. For example, in Chen et al. (2024), the authors studied the effect of two types of medication on blood clot dissolution. An unmea-

[16]Related work by Gaebler et al. (2024) has targeted a similar question although their study does not explore how the model's beliefs vary with race and the subsequent impact of this on the decision as the object of inference.

[17]Of course, such a document is subject to a host of measurement error issues, such as memory affecting recollection of the incident and fear of repercussions due to misconduct, although, this entire enterprise is not without its challenges.

sured confounder preventing a clean causal contrast is the presence of atrial fibrilation. For each patient, they take two distinct pieces of pre-treatment clinical text, for example, an echocardiogram (ECG) note and a nursing note. They then ask an LLM, zero-shot, whether the note suggests that the patient has the unobserved confounding condition (the LLM must respond with Yes or No). By doing this again with another LLM, under certain conditions, the authors leverage human output—the nursing note and ECG—to straightforwardly generate two independent proxy measures of atrial fibrilation, which they can use to correct the confounding bias. Innovative ideas such as these I believe lie ahead for the identification of causal effects in observational data where beliefs are present.

Alternative identification strategies may exploit restrictions in the data-generating process to remove the effect of certain problematic mechanisms. An experimental design, such as the parallel encouragement design (Imai et al., 2013), is one example, where indirect and direct effects can be reliably separated without sequential exchangeability. For example, consider an experimental hiring study in which evaluators are asked to make interview or hiring recommendations for otherwise identical applicants. Applicant race could be randomized through names or photographs, while a separate randomized encouragement provides information designed to shift the evaluator's belief about future job performance—for instance, an externally generated assessment indicating that the applicant is likely to perform well. Other features could also be manipulated, such as those indicating the education of the applicant. The design could then distinguish changes in hiring caused by changes in racially informed perceived productivity from changes in hiring that persist despite experimentally induced variation in those beliefs. This is not to say that studies evaluating the existence of statistical discrimination do not exist, however, this design provides a method for identifying (inaccurate) statistical discrimination as it is defined within my proposed framework.

Another strategy might exploit a hypothesized absence of taste-based discrimination in observed data, so that any remaining discrimination is statistical (whether or not it is accurate). Returning to the territory of the Latin Kings as in Section 3, suppose that

this neighborhood is policed by an all-White force. Suppose further that its residents are either Black or Hispanic. One might hypothesize that officers do not exhibit a taste for discrimination against a particular group. Therefore, finding discrimination in this context strongly indicates statistical discrimination is occurring. Validating this hypothesis with data would be challenging, but in collaboration with a qualitative analysis of departmental operations, a more robust empirical treatment could be offered.[18]

8 Conclusion

In this article, I provide the first fully causal treatment of the outcome test for discrimination. I introduce a graphical notion of the beliefs of the decision-maker that is derived from the causal model without conceptual loss of generality. Using this model, I define novel and model agnostic notions of taste-based and statistical discrimination. I review the traditional implementation of the outcome, unveiling a general treatment of the hit rate test that blueprints a set of assumptions required for it to give a valid inference of discrimination. I discuss the marginal outcome test and its reliance on an assumption regarding calibrated beliefs of the decision-maker. Leveraging a new appreciation of the test using ideas from proximal causal inference, I derive new bounds on several causal estimands seldom specified in uses of the outcome test, restoring the intuitive meaning of a hit rate disparity. I apply these results to an empirical study of traffic stops and searches in Washington state, where I find notable evidence of racial bias in searches against Black and Hispanic drivers. I conclude with a discussion of the implications of my causal model for future work.

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[18] I would like to thank Dean Knox for introducing me to the (existence of the) Latin Kings and Aaron Kaufman for a discussion which helped me refine this idea.

debugging, assistance with figures and plots, and checking the consistency of all mathematical arguments. All AI-assisted outputs, including text, code, analyses, and figures, were reviewed, verified, and edited by the author, who remains responsible for the accuracy, validity, and final content of the manuscript.

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A Proofs

A.1 Proof of Proposition 1

Proof. For part (i), fix race r and officer type u_d. Since D is binary, the map $b \mapsto f_D(r, b, u_d)$ partitions $\mathcal{B}$ into the no-search region $\mathcal{B}_0(r, u_d) := \{b : f_D(r, b, u_d) = d_0\}$ and the search region $\mathcal{B}_1(r, u_d) := \{b : f_D(r, b, u_d) = d_1\}$. Although the partition may vary with u_d the collection of all such partitions jointly partitions $\mathcal{B}$ into cells, where two belief states occupy the same cell exactly when they induce the same decision almost surely for every realization of U_D. This is the equivalence relation $\sim_r$.

The resulting variable $B^\dagger(r) = [B(r)]_{\sim_r}$ therefore retains exactly the distinctions among belief states that can matter for the decision under race r. In particular, $f_D(r, \cdot, u_d)$ is almost surely constant within each equivalence class, so we may define $f_D^\dagger(r, [b]_{\sim_r}, u_d) := f_D(r, b, u_d)$. This is well-defined by construction and yields (3). Thus, replacing $B(r)$ by $B^\dagger(r)$ entails no loss for the decision mechanism. Doing so for each r means we may replace factual B with $B^\dagger$.

For part (ii), associate with every canonical belief $b^\dagger$ the set $A_r(b^\dagger)$ of types for which it induces search. Define $b^\dagger \preceq_r b^{\dagger\prime}$ whenever $A_r(b^\dagger) \subseteq A_r(b^{\dagger\prime})$ almost surely. Under this ordering, almost every officer type that searches at $b^\dagger$ also searches at $b^{\dagger\prime}$.

The relation $\preceq_r$ is a total order on the canonical belief space. Reflexivity and transitivity follow from set inclusion. For antisymmetry, if both $b^\dagger \preceq_r b^{\dagger\prime}$ and $b^{\dagger\prime} \preceq_r b^\dagger$, then $A_r(b^\dagger) = A_r(b^{\dagger\prime})$ almost surely. Hence $f_D^\dagger(r, b^\dagger, U_D) = f_D^\dagger(r, b^{\dagger\prime}, U_D)$ almost surely, and part (i) implies $b^\dagger = b^{\dagger\prime}$. Completeness follows directly from the nesting assumption, which requires one of $A_r(b^\dagger)$ and $A_r(b^{\dagger\prime})$ to be contained almost surely in the other.

Equipped with the total order, it follows that there exists an ordered scalar representation of $B^\dagger$. Indeed, choosing

$$\phi_r(b^\dagger) := \mathbb{P}\big(U_D \in A_r(b^\dagger)\big) = \mathbb{P}\Big(f_D^\dagger(r, b^\dagger, U_D) = d_1\Big), \tag{24}$$

we have by construction, $\phi_r(b^\dagger) \in [0, 1]$. If $b^\dagger \preceq_r b^{\dagger\prime}$, the nesting condition implies $\phi_r(b^\dagger) \leq \phi_r(b^{\dagger\prime})$. Moreover, if $b^\dagger \neq b^{\dagger\prime}$, part (i) implies that their officer types differ on a set of U_D-values having positive probability. Since the corresponding sets are nested, their probabilities must then differ strictly. Hence $b^\dagger \prec_r b^{\dagger\prime}$ implies $\phi_r(b^\dagger) < \phi_r(b^{\dagger\prime})$, so ϕ_r is an injective, order-preserving scalar representation.

Let $\widetilde{B}(r) := \phi_r(B^\dagger(r))$. Fix u_d and suppose $\phi_r(b^\dagger) < \phi_r(b^{\dagger\prime})$. If $f_D^\dagger(r, b^\dagger, u_d) = d_1$, then $u_d \in A_r(b^\dagger)$. Since ϕ_r is preserves order, $A_r(b^\dagger) \subseteq A_r(b^{\dagger\prime})$ almost surely, and therefore $u_d \in A_r(b^{\dagger\prime})$ for almost every u_d. It follows that the binary decision is weakly increasing

in $\widetilde{B}(r)$ almost surely. The region of $\widetilde{B}(r)$ which induces a search is consequently an upper interval of the support of $\widetilde{B}(r)$, so there exists a threshold $t(r, u_d)$ yielding (4). ■

A.2 Proof of Proposition 2

Proof. Conditional on $(D = d_1, R = r_1)$, only AS and PF encounters are possible, whereas conditional on $(D = d_1, R = r_0)$, only AS and AW are possible. Using the law of total probability and consistency, these facts imply that

$$\begin{aligned}\mathbb{E}[Y \mid D = d_1, R = r_1] = \mathbb{E}[Y \mid \text{AS}, R = r_1] + \\ \Pr(\text{PF} \mid D = d_1, R = r_1)\{\mathbb{E}[Y \mid \text{PF}, R = r_1] - \mathbb{E}[Y \mid \text{AS}, R = r_1]\},\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[Y \mid D = d_1, R = r_0] = \mathbb{E}[Y \mid \text{AS}, R = r_0] + \\ \Pr(\text{AW} \mid D = d_1, R = r_0)\{\mathbb{E}[Y \mid \text{AW}, R = r_0] - \mathbb{E}[Y \mid \text{AS}, R = r_0]\}.\end{aligned}$$

Subtracting the second identity from the first gives $\Delta = (1) + (2) - (3)$.

For the equivalent representations as covariances, set $Z = D(r_1) - D(r_0)$. Given $(D = d_1, R = r_1)$, Z is binary: it equals one on PF and zero on AS. By definition of covariance, then

$$\begin{aligned}\text{Cov}(Y(d_1), Z \mid D = d_1, R = r_1) = \Pr(\text{PF} \mid D = d_1, R = r_1) \Pr(\text{AS} \mid D = d_1, R = r_1) \\ \times \{\mathbb{E}[Y \mid \text{PF}, R = r_1] - \mathbb{E}[Y \mid \text{AS}, R = r_1]\},\end{aligned}$$

which yields the stated expression for (2). Given $(D = d_1, R = r_0)$, Z equals minus one on AW and zero on AS, so the same calculation gives

$$\begin{aligned}\text{Cov}(Y(d_1), Z \mid D = d_1, R = r_0) = -\Pr(\text{AW} \mid D = d_1, R = r_0) \Pr(\text{AS} \mid D = d_1, R = r_0) \\ \times \{\mathbb{E}[Y \mid \text{AW}, R = r_0] - \mathbb{E}[Y \mid \text{AS}, R = r_0]\}.\end{aligned}$$

Dividing by the corresponding always-searched probability gives the stated expression for (3). ■

A.3 Proof of Proposition 3

Proof. Let $q_r = \Pr(Y(d_1) = 1 \mid R = r)$, $a_r = \Pr(D = d_1 \mid R = r, Y(d_1) = 1)$, and $b_r = \Pr(D = d_1 \mid R = r, Y(d_1) = 0)$, assumed all strictly positive. Clairvoyance and consistency imply that the hit rate among searched motorists of race r is $h_r = q_r a_r / [q_r a_r + (1 - q_r) b_r]$. Because R and $Y(d_1)$ are binary, condition (i) implies that $q_{r_1} > q_{r_0}$; similarly, the observation that $\mathrm{Cov}(Y, R \mid D = d_1)$ implies that $h_{r_1} < h_{r_0}$.

Study the odds of recovering contraband, where

$$\frac{h_r}{1 - h_r} = \frac{q_r}{1 - q_r} \frac{a_r}{b_r}.$$

The above display reveals that $q_{r_1} > q_{r_0}$ together with $h_{r_1} < h_{r_0}$ requires $a_{r_1}/b_{r_1} < a_{r_0}/b_{r_0}$. By condition (ii), $\delta_r := a_r - b_r \geq 0$, and condition (iii) states that $\delta_{r_1} > \delta_{r_0}$, which implies that $\delta_{r_1}/b_{r_1} < \delta_{r_0}/b_{r_0}$. Consequently $b_{r_1}/b_{r_0} > \delta_{r_1}/\delta_{r_0} > 1$, so $b_{r_1} > b_{r_0}$.

Thus, among motorists with $Y(d_1) = 0$, minority status raises the search propensity. In the stylized model of the proposition, where we assume officer clairvoyance, $Y(d_1)$ is sufficient to control for confounding of the effect $R \to D$. Therefore $b_{r_1} - b_{r_0} = \mathbb{E}[D(r_1) - D(r_0) \mid Y(d_1) = 0] > 0$. A positive average contrast implies that $D(r_1) > D(r_0)$ on a set of positive probability, establishing racial bias in the sense of Definition 1. ■

A.4 Proof of Proposition 5

Proof. Suppress X for brevity. Write $h_r(y) = \Pr(Y = y \mid D = d_1, R = r)$, such that $h_r(1)$ is the hit rate for racial group r, and $h_r(0)$ is the corresponding miss rate. Under Assumption 1 (monotonicity), a searched White motorist must belong to AS, while a searched minority motorist belongs to either AS or PF. Hence, the probability of necessity can be equivalently written $\theta = \Pr(\mathrm{PF} \mid D = d_1, R = r_1)$ and, similarly, by consistency and Assumption 2, $h_{r_0}(y) = \Pr\{Y(d_1) = y \mid \mathrm{AS}, R = r_1\}$ for each $y \in \{0, 1\}$.

Let $m_y = \Pr(Y(d_1) = y \mid \mathrm{PF}, R = r_1)$. The minority hit rate is $h_{r_1}(y) = (1 - \theta) h_{r_0}(y) + \theta m_y$. Since $m_y \geq 0$, $h_{r_1}(y) \geq (1 - \theta) h_{r_0}(y)$ for each outcome value with $h_{r_0}(y) > 0$. Thus $\theta \geq 1 - h_{r_1}(y)/h_{r_0}(y)$ for each y. Taking the largest of these two lower bounds is equivalent to subtracting the smaller ratio from one, which gives the lower endpoint in (16). The upper bound $\theta \leq 1$ follows because θ is a probability.

Next, I show the bounds are sharp. Assume causal model (1) which contains model (2) without loss, via Proposition 1. Note that there are no conditional independence restrictions we can leverage here, so we reason about the principal strata. Generate a DGP in which the lower bound is attained, where, if minorities have the lower hit rate, set discriminatory

searches to be entirely unproductive, so that $m_1 = 0$, and do the reverse if minorities have a higher hit rate. The construction above shows that $\theta = \underline{\theta}$ in these cases. When the hit rates are equal, $\underline{\theta} = 0$, and the lower endpoint is attained by assigning all searched minority motorists to the AS stratum, so that $\theta = 0$. In other words, in a world where no minority motorists were profiled in their search. The upper bound is attained by making all minority searches discriminatory (all searches are profiled searches PF), which the stated assumptions do not prohibit. ■

A.5 Proof of Proposition 6

Proof. Suppress X for brevity and continue to write $h_r(y) = \Pr(Y = y \mid D = d_1, R = r)$. Let $\lambda = \Pr(\text{AW} \mid D = d_1, R = r_0)$, $a_r(y) = \Pr\{Y(d_1) = y \mid \text{AS}, R = r\}$, and $b_y = \Pr\{Y(d_1) = y \mid \text{AW}, R = r_0\}$. By consistency, a searched White motorist belongs to either AS or AW, and hence

$$h_{r_0}(y) = (1 - \lambda)a_0(y) + \lambda b_y. \tag{25}$$

Since $b_y \leq 1$ and $\lambda \leq \eta$, it follows that

$$a_0(y) \geq \max\left\{0, \frac{h_{r_0}(y) - \eta}{1 - \eta}\right\}. \tag{26}$$

The restriction $(1 - \rho)a_0(1) \leq a_1(1) \leq a_0(1)$ therefore implies

$$a_1(1) \geq (1 - \rho)\max\left\{0, \frac{h_{r_0}(1) - \eta}{1 - \eta}\right\}, \qquad a_1(0) \geq \max\left\{0, \frac{h_{r_0}(0) - \eta}{1 - \eta}\right\}. \tag{27}$$

The second inequality follows because $a_1(1) \leq a_0(1)$ is equivalent to $a_1(0) \geq a_0(0)$.

By consistency, a searched minority motorist belongs to either AS or PF, irrespective of whether monotonicity holds. Consequently, $\theta = \Pr(\text{PF} \mid D = d_1, R = r_1)$. Letting $m_y = \Pr\{Y(d_1) = y \mid \text{PF}, R = r_1\}$ gives

$$h_{r_1}(y) = (1 - \theta)a_1(y) + \theta m_y \geq (1 - \theta)a_1(y). \tag{28}$$

Combining (27) and (28) gives the two nontrivial lower bounds appearing in (17); the constant one inside the minimum records the additional restriction $\theta \geq 0$. Their maximum is $\underline{\theta}_{\eta,\rho}$, while $\theta \leq 1$ follows because θ is a probability.

For sharpness, first suppose the ratio of hit rates attains the minimum in (17). Set the anti-White searches to be entirely successful, choose the always-searched White hit rate at its lower feasible value in (26), set the always-searched minority hit rate equal to $(1 -$

$\rho)a_0(1)$, and make profiled searches entirely unsuccessful. These choices attain the lower endpoint. If the miss-rate ratio attains the minimum, reverse the outcomes assigned to the anti-White and profiled strata and set $a_1(1) = a_0(1)$; this again attains the lower endpoint. Finally, if the constant one is the minimum, the observed minority outcome distribution lies within the permitted range for the always-searched minority distribution, so all searched minority motorists may be assigned to AS and $\theta = 0$. The remaining principal-stratum probabilities are unrestricted and may be chosen to reproduce the observed distribution. As in Proposition 5, the upper endpoint is attained by assigning all minority searches to PF. ■

Next I derive the bound when Assumption 2 is relaxed in the opposite direction.

Proposition 7 *Fix $\eta \in [0, 1)$ and $\kappa \geq 1$. Suppose that $\Pr(\text{AW} \mid D = d_1, R = r_0) \leq \eta$ and that (19) holds. Then the sharp identified set for θ is $[\underline{\theta}_{\eta,\kappa}, 1]$, where*

$$\underline{\theta}_{\eta,\kappa} = 1 - \min\left\{1,\ \frac{(1-\eta)\, h_{r_1}(1)}{h_{r_0}(1) - \eta},\ \frac{(1-\eta)\, h_{r_1}(0)}{1 - \eta - \kappa\, h_{r_0}(1)}\right\}, \tag{29}$$

with the convention that a term of the minimum whose denominator is nonpositive equals $+\infty$. Setting $\kappa = 1$ returns the bound of Proposition 6 evaluated at $\rho = 0$.

Proof. Suppress X for brevity and continue to write $h_r(y) = \Pr(Y = y \mid D = d_1, R = r)$. Let $\lambda = \Pr(\text{AW} \mid D = d_1, R = r_0)$, $a_r(y) = \Pr\{Y(d_1) = y \mid \text{AS}, R = r\}$, and $b_y = \Pr\{Y(d_1) = y \mid \text{AW}, R = r_0\}$. By consistency, a searched White motorist belongs to either AS or AW, and hence

$$h_{r_0}(y) = (1 - \lambda)a_0(y) + \lambda b_y. \tag{30}$$

Since $b_1 \leq 1$ and $\lambda \leq \eta$,

$$a_0(1) \geq \max\left\{0, \frac{h_{r_0}(1) - \eta}{1 - \eta}\right\}, \tag{31}$$

while discarding the nonnegative term λb_1 in (30) and using $1 - \lambda \geq 1 - \eta$ gives

$$a_0(1) \leq \min\left\{1, \frac{h_{r_0}(1)}{1 - \eta}\right\}. \tag{32}$$

The restriction $a_0(1) \leq a_1(1) \leq \kappa a_0(1)$ therefore implies

$$a_1(1) \geq \max\left\{0, \frac{h_{r_0}(1) - \eta}{1 - \eta}\right\}, \qquad a_1(0) \geq \max\left\{0,\ 1 - \kappa \min\left\{1, \frac{h_{r_0}(1)}{1 - \eta}\right\}\right\}. \tag{33}$$

The first inequality uses $a_1(1) \geq a_0(1)$ together with (31) and does not involve κ; the second uses $a_1(0) = 1 - a_1(1) \geq 1 - \kappa a_0(1)$ together with (32).

By consistency, a searched minority motorist belongs to either AS or PF, irrespective of whether monotonicity holds. Consequently, $\theta = \Pr(\text{PF} \mid D = d_1, R = r_1)$. Letting $m_y = \Pr\{Y(d_1) = y \mid \text{PF}, R = r_1\}$ gives

$$h_{r_1}(y) = (1-\theta)a_1(y) + \theta m_y \geq (1-\theta)a_1(y). \tag{34}$$

Combining (33) and (34) gives the two nontrivial lower bounds appearing in (29); the constant one inside the minimum records the additional restriction $\theta \geq 0$. Their maximum is $\underline{\theta}_{\eta,\kappa}$, while $\theta \leq 1$ follows because θ is a probability. When the lower bound on $a_1(0)$ in (33) equals zero, which occurs precisely when $1 - \eta - \kappa h_{r_0}(1) \leq 0$, the corresponding inequality in (34) restricts θ not at all; this is the convention that the final term of the minimum is $+\infty$.

For sharpness, first suppose the ratio of hit rates attains the minimum in (29). Set $\lambda = \eta$ and the anti-White searches to be entirely successful, so that $a_0(1)$ takes its lower feasible value in (31); set the always-searched minority hit rate equal to $a_0(1)$, the lower end of the band permitted by (19); and make profiled searches entirely unsuccessful. These choices attain the lower endpoint. If the miss-rate ratio attains the minimum, reverse the outcomes assigned to the anti-White and profiled strata, so that $a_0(1)$ takes its upper feasible value in (32), and set $a_1(1) = \kappa a_0(1)$; this again attains the lower endpoint. Finally, if the constant one is the minimum, then $h_{r_1}(1)$ is at least the right-hand side of (31) and at most κ times the right-hand side of (32), so the interval

$$\left[\max\left\{\frac{h_{r_1}(1)}{\kappa},\, \max\left\{0,\, \frac{h_{r_0}(1)-\eta}{1-\eta}\right\}\right\},\; \min\left\{h_{r_1}(1),\, \min\left\{1,\, \frac{h_{r_0}(1)}{1-\eta}\right\}\right\}\right]$$

is nonempty. Choosing $a_0(1)$ within it and setting $a_1(1) = h_{r_1}(1)$ satisfies (19), so all searched minority motorists may be assigned to AS and $\theta = 0$. The remaining principal-stratum probabilities are unrestricted and may be chosen to reproduce the observed distribution. As in Proposition 5, the upper endpoint is attained by assigning all minority searches to PF. ■

B Miscalibration and Identification

B.1 Identification via Bridge Functions

A fundamental difficulty in this setting is that $Y(d_1)$ is binary while V may be high-dimensional. The proxy therefore contains too little variation to distinguish all of the circumstances that govern the search decision, a failure commonly described as a lack of

completeness. An unsuccessful inventory search and an unproductive furtive search may both have $Y(d_1) = 0$, even though the former occurs regardless of race and the latter occurs only when the motorist is perceived as minority. Similarly, a fragrant search and a courier may both have $Y(d_1) = 1$, despite belonging to different principal strata. A hit or miss alone cannot generally distinguish the circumstances that determine whether, and for whom, a search occurs.

Some work addresses partial identification of the counterfactual mean without completeness by instead assuming the existence of a bridge function. In this setting, a bridge is a function h satisfying

$$\mathbb{E}[D \mid R, V] = \mathbb{E}[h(R, Y(d_1)) \mid R, V]. \tag{35}$$

In words, (35) requires some transformation of $Y(d_1)$ to be sufficiently informative about V to reproduce the officer's search propensity by race for every value of V. Because V is sufficient for confounding control, $\mathbb{E}[D \mid R, V] = \mathbb{E}[D(r) \mid V]$, so such a bridge would make the proxy directly informative about discrimination as defined in Definition 1. The inability of the binary outcome to distinguish these motorists illustrates why this is a demanding requirement.

The restriction becomes sharper when officer beliefs are represented explicitly. Recall that, without loss, V may be represented by $(B(r_0), B(r_1), Y(d_1))$. Equation (35) then reduces to

$$\mathbb{E}[D \mid R, B(r_0), B(r_1), Y(d_1)] = h(R, Y(d_1)), \tag{36}$$

so that after conditioning on the officer's potential beliefs, the search propensity must still be representable using only race and the potential discovery of contraband. This resembles a strong form of calibration: beliefs must be sufficiently tied to $Y(d_1)$ for the binary proxy to stand in for the contextual information that drives search. Officer clairvoyance, $B = Y(d_1)$, is one condition sufficient for this restriction.

Bridge conditions are typically difficult to interpret, making potential violations challenging to interrogate. Here the examples show concretely why the condition may fail. The identification strategy developed above therefore does not rely on a bridge condition, in contrast to some recent developments in this area (Ghassami et al., 2026).

B.2 Incorporating Misclassification into Proposition 5

Building on the introduction of beliefs in Section 2, I now extend Proposition 5 to account for how estimates of discrimination are sensitive to the misspecification of officer beliefs. This purpose of this exercise is to explore possible future directions in theory building for partial identification when officer beliefs are misspecified.

Recall that (cf Section 2) officers operate without loss of generality under the threshold rule

$$D(r, B(r, V), U_D) = \mathbf{1}\{B(r, V) > t(r, U_D)\}. \tag{37}$$

Consider the success rate of a minority search that was discriminatory, $\mathbb{E}[Y(d_1) \mid \text{PF}, R = r_1]$. Assume that among these searches, officers on average overestimate contraband risk by at most ε such that

$$\mathbb{E}[B(r_1) - Y(d_1) \mid \text{PF}, R = r_1] \leq \varepsilon. \tag{38}$$

Since for every profiled search, $D(r_1) = 1 \iff B(r_1) > t(r_1)$, it follows that $\mathbb{E}[B(r_1) \mid \text{PF}, R = r_1] \geq \mathbb{E}[t(r_1) \mid \text{PF}, R = r_1] =: \bar{t}_1$. From this it follows that

$$\mathbb{E}[Y(d_1) \mid \text{PF}, R = r_1] \geq \max\{0, \bar{t}_1 - \varepsilon\}. \tag{39}$$

To restate the above, the average evidential standard officers set during discriminatory searches of minorities I denote by $\bar{t}_1$. If officers are permitted to oversuspect minorities by at most ε, the success of a discriminatory search is then lower bounded by their difference, $\bar{t}_1 - \varepsilon$, when this difference is positive.

Suppose that, on average, officers overestimate minority contraband risk by $\varepsilon = 15$ percentage points. If officers require minorities to have a perceived contraband risk of only $\bar{t}_1 = 10\%$ before searching, then this restriction places no positive lower bound on the success rate of discriminatory searches. The reason is that officers' misperceptions are sufficiently poor, relative to their willingness to search, that even motorists with zero actual probability of carrying contraband may nevertheless appear (to the officer) sufficiently risky to be searched. Thus, the fact that a discriminatory search occurs may not imply anything about underlying contraband risk. Now suppose instead that officers overestimate minority contraband risk by at most $\varepsilon = 5$ percentage points, while their average willingness to search remains $\bar{t}_1 = 10\%$. In this case, miscalibrated beliefs alone are not enough to push a motorist with zero contraband risk above the search threshold. At most five percentage points of the perceived risk needed to justify the search can be attributed to officers' overestimation, so the remaining five percentage points must be informed by actual contraband risk, i.e. some officers were definitely on to something. The success rate of discriminatory searches must therefore be at least $\bar{t}_1 - \varepsilon = 5\%$. These examples explain the relation in (39).

Leveraging these ideas, I derive a corollary of Proposition 5.

Proposition 8 *Let* $\bar{t}_1 \coloneqq \mathbb{E}[t(r_1, U_D) \mid \text{PF}, R = r_1]$ *be the average willingness to search among discriminatory searches of minorities. In addition to Assumptions 1 and 2, assume that* $\mathbb{E}[B(r_1) - Y(d_1) \mid \text{PF}, R = r_1] \leq \varepsilon$, *for some* $\varepsilon \in [0, 1]$. *Then, whenever* $\mathbb{E}[Y \mid D =$

$d_1, R = r_0] > \max\{0, \bar{t}_1 - \varepsilon\}$,

$$\theta \in \left[\frac{\mathbb{E}[Y \mid D = d_1, R = r_0] - \mathbb{E}[Y \mid D = d_1, R = r_1]}{\mathbb{E}[Y \mid D = d_1, R = r_0] - \max\{0, \bar{t}_1 - \varepsilon\}},\ 1 \right], \tag{40}$$

Proof. In the exposition above this proposition, it is shown in (39) that $\mathbb{E}[Y(d_1) \mid \text{PF}, R = r_1] \geq \max\{0, \bar{t}_1 - \varepsilon\}$. Write $c \coloneqq \max\{0, \bar{t}_1 - \varepsilon\}$ for this lower bound on the success rate of a discriminatory search, and let $h_{r_0} = \mathbb{E}[Y \mid D = d_1, R = r_0]$ and $h_{r_1} = \mathbb{E}[Y \mid D = d_1, R = r_1]$ be the White and minority hit rates respectively. By Assumptions 1 and 2, the same argument in Proposition 5 gives $h_{r_1} = (1 - \theta)h_{r_0} + \theta\mathbb{E}[Y(d_1) \mid \text{PF}, R = r_1]$. Since $\mathbb{E}[Y(d_1) \mid \text{PF}, R = r_1] \geq c$, $h_{r_1} \geq h_{r_0} - \theta(h_{r_0} - c)$. When $h_{r_0} > c$, rearranging yields $\theta \geq (h_{r_0} - h_{r_1})/(h_{r_0} - c)$, which is the lower endpoint in (40). The upper bound again follows from $\theta \leq 1$.[19] ■

It is instructive to separate the roles of $\bar{t}_1$ and ε into the types of behavioral mechanism they represent. Indeed, echoing Definition 2, $\bar{t}_1$ describes the average 'taste' officers possess to search a vehicle during a discriminatory stop of a minority. Relatedly, ε describes the impacts of inaccurate statistical discrimination, among searches of minority drivers where race was used as a factor in the decision, ε describes the maximum average error officers were making in predicting contraband risk. It is expected that these mechanisms manifest inextricably in the bound for θ, since, as discussed in Section 2.1, their observable implications can be identical.

Inspecting (40), find that among discriminatory searches as officer standards for searching minorities increases (holding ε fixed), the lower bound on θ increases, indicating a greater prevalence of discrimination. Why does this happen? Note that an increasing $\bar{t}_1$ in this case does not indicate a reversal in racial preferences for search because it conditions on the set of discriminatory searches. For a given observed disparity in the hit rate, increasing standards for searching minorities—in encounters where a White driver would not have been searched—likely entails the discovery of more contraband. Therefore, one can only reconcile the observation that $\mathbb{E}[Y \mid D = d_1, R = r_1] \ll \mathbb{E}[Y \mid D = d_1, R = r_0]$ with a large prevalence of discriminatory searches.

Similarly, the more officers oversuspect minority drivers (holding $\bar{t}_1$ fixed), the smaller the lower bound on θ, indicating less discrimination. This is because oversuspicion of minority drivers among discriminatory searches leads to a lower productivity of those searches. Therefore, the observation that $\mathbb{E}[Y \mid D = d_1, R = r_1] \ll \mathbb{E}[Y \mid D = d_1, R = r_0]$ is reconciled

[19] On sharpness, this bound is only sharp when minority hit rates do not exceed White hit rates, following a similar argument to earlier results. As the purpose of this exercise is to build intuition, I do not consider both cases for clarity.

with a smaller prevalence of discriminatory searches.

Separately, these mechanisms correspond to the mediating pathways of Section 2.1. However, it is how they balance in tandem which ultimately affects the bound for θ. A large lower bound, explained by a large difference $\bar{t}_1 - \varepsilon$, could be explained by, among discriminatory searches of minorities, stringent thresholds for search, calibrated predictions of contraband discovery or some antagonistic combination of the two. The intuition for this result is closely tied to the notion of a productive discriminatory search and its relationship with the beliefs of the officer, see Section 3.

Now, it is generally difficult to motivate a choice for ε and $\bar{t}_1$ in Proposition 8 to use in some empirical analysis. I use this result as a proof of concept. The above discussion reveals that theory building on how the productivity of discrimination is perceived by officers could lead to tighter bounds on discrimination, making better use of the data typically available to researchers.

C Estimation details

In keeping with the proofs of Appendix A, throughout this section, write $h_r(y \mid x) = \Pr(Y = y \mid D = d_1, R = r, X = x)$, so that $h_r(1 \mid x)$ is the hit rate of racial group r in subpopulation x and $h_r(0 \mid x)$ the corresponding miss rate. In this notation the quantity reported in Section 6 is the lower endpoint of Proposition 5 evaluated within a subpopulation,

$$\underline{\theta}(x) = 1 - \min_{y \in \{0,1\}} \frac{h_{r_1}(y \mid x)}{h_{r_0}(y \mid x)}, \tag{41}$$

with the unconditional bound $\underline{\theta}$ the same expression formed from the pooled rates. Let $n_r(x)$ denote the number of searched motorists of race r observed in subpopulation x, and $n_r(y, x)$ the number of those whose search returned outcome y. I estimate the rates by their empirical counterparts, $\hat{h}_r(y \mid x) = n_r(y, x)/n_r(x)$, giving

$$\hat{\underline{\theta}}(x) = 1 - \min_{y \in \{0,1\}} \frac{\hat{h}_{r_1}(y \mid x)}{\hat{h}_{r_0}(y \mid x)}. \tag{42}$$

Writing $q_y(x) = h_{r_1}(y \mid x)/h_{r_0}(y \mid x)$, the conditional bound is $\underline{\theta}(x) = 1 - \phi(q_1(x), q_0(x))$ with $\phi(a, b) = \min(a, b)$. This map is non-regular due to the lack of differentiability on the

diagonal $a = b$. It is Hadamard *directionally* differentiable, though, with

$$\phi'_{(a,b)}(u,v) = \begin{cases} u, & a < b, \\ v, & b < a, \\ \min(u,v), & a = b, \end{cases} \tag{43}$$

and for functionals of this kind the standard bootstrap is inconsistent (Dümbgen, 1993; Fang and Santos, 2019). Resampling the data and re-evaluating ϕ within each replication does not produce a consistent estimator of the sampling distribution if (a, b) lies on or near the diagonal non-trivially often. Whether that matters in a given application depends on how close the estimated ratios lie to the diagonal. For inference I use the numerical delta method of Fang and Santos (2019), which replaces the re-evaluation of ϕ by its estimated directional derivative, which has been shown to restore consistency of the bootstrap.

Let n denote the number of searched motorists in the pair being compared. For each of B replications I resample these motorists with replacement, recompute the rates $\hat{h}^*_r(y \mid x)$, and form the perturbations $u_r(y,x) = \sqrt{n}\{\hat{h}^*_r(y \mid x) - \hat{h}_r(y \mid x)\}$. Propagating these through q_y by the ordinary delta method gives

$$\dot{q}_y(x) = \frac{u_{r_1}(y,x)}{\hat{h}_{r_0}(y \mid x)} - \frac{\hat{h}_{r_1}(y \mid x)\, u_{r_0}(y,x)}{\hat{h}_{r_0}(y \mid x)^2}, \tag{44}$$

and the replication contributes

$$T^* = -\,\hat{\phi}'\big(\dot{q}_1(x), \dot{q}_0(x)\big). \tag{45}$$

This is due entirely to the non-regularity in ϕ. Because the estimand is not an average over X, there is no accompanying delta-method contribution from an estimated covariate distribution.

The derivative appearing in (45) is estimated by classifying each cell according to $\hat{q}_1(x) - \hat{q}_0(x)$ against a tolerance $\tau_n = \sqrt{\log n}/\sqrt{n}$, applying the first branch of (43) when $\hat{q}_1(x) < \hat{q}_0(x) - \tau_n$, the second when $\hat{q}_0(x) < \hat{q}_1(x) - \tau_n$, and the third otherwise. The tolerance satisfies $\tau_n \to 0$ and $\sqrt{n}\,\tau_n \to \infty$, which is required for consistency of the estimated derivative. In the small number of cells where a denominator in (44) vanishes I substitute the numerical difference $\sqrt{n}\{\hat{\underline{\theta}}^*(x) - \hat{\underline{\theta}}(x)\}$ for that cell.

Accommodating the sensitivity parameters of Proposition 6 requires only three changes and no new machinery. First, the ratios entering the minimum are replaced by their adjusted counterparts, $q_1^{\eta,\rho}(x) = \{(1-\eta)/(1-\rho)\}\,\hat{h}_{r_1}(1 \mid x)/\{\hat{h}_{r_0}(1 \mid x) - \eta\}$ and $q_0^{\eta,\rho}(x) = (1-\eta)\,\hat{h}_{r_1}(0 \mid$

$x)/\{\hat{h}_{r_0}(0 \mid x) - \eta\}$, with a ratio whose denominator is nonpositive set to $+\infty$ and thereby ignored. Second, the perturbation in (44) is modified correspondingly, the White rate in each denominator becoming $\hat{h}_{r_0}(y \mid x) - \eta$ and the whole expression scaled by $1 - \eta$, together with $1/(1-\rho)$ for $y = 1$. Third, as minimum is taken over three candidates rather than two, its derivative is adjusted accordingly, however this is not a major alteration because when the constant unity binds, the derivative vanishes. Everything else is unchanged and at $(\eta, \rho) = (0, 0)$ the procedure returns the estimates of Proposition 5 exactly.

Because η and ρ are fixed constants assumed known, they contribute no additional sampling variability. One practical remark: the bootstrap resamples are drawn once per comparison and reused at every (η, ρ), so that differences across the sensitivity curves reflect the parameters alone and not simulation error. This is what makes the curves in Figures 6 and 7 pointwise comparable.

The empirical distribution of $\{T_b^*\}_{b=1}^B$ approximates that of $\sqrt{n}(\hat{\underline{\theta}} - \underline{\theta})$. Denoting by c_α^* its α quantile, I report a one-sided lower confidence limit $\hat{\underline{\theta}} - c_{1-\alpha}^*/\sqrt{n}$, truncated below at zero, which covers the interval $[\underline{\theta}, 1] \ni \theta$ with coverage rate 95%.

D Additional Tables and Figures for Sensitivity Analysis

D.1 Additional Figures

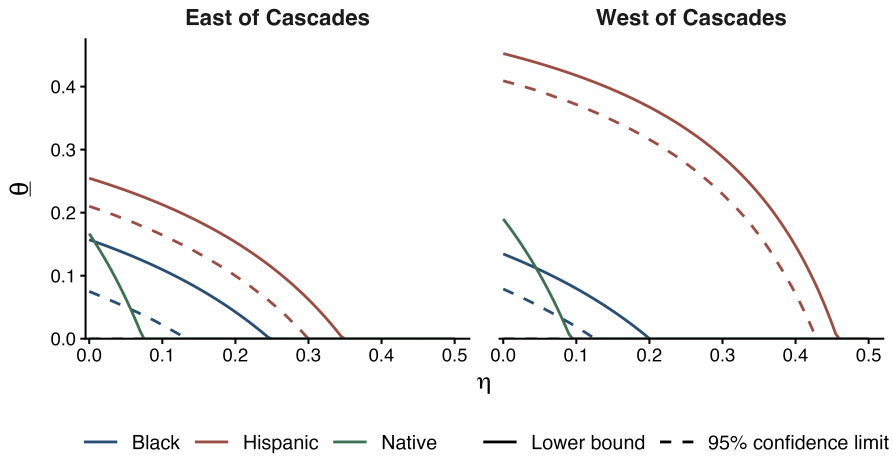


Figure 8: **Sensitivity to violations of Assumption 1, by side of the Cascade crest.** Solid lines are $\hat{\underline{\theta}}_{\eta,0}$ and dashed lines the one-sided 95% confidence limit. The confidence limit for Native motorists is zero throughout both panels and so coincides with the horizontal axis.

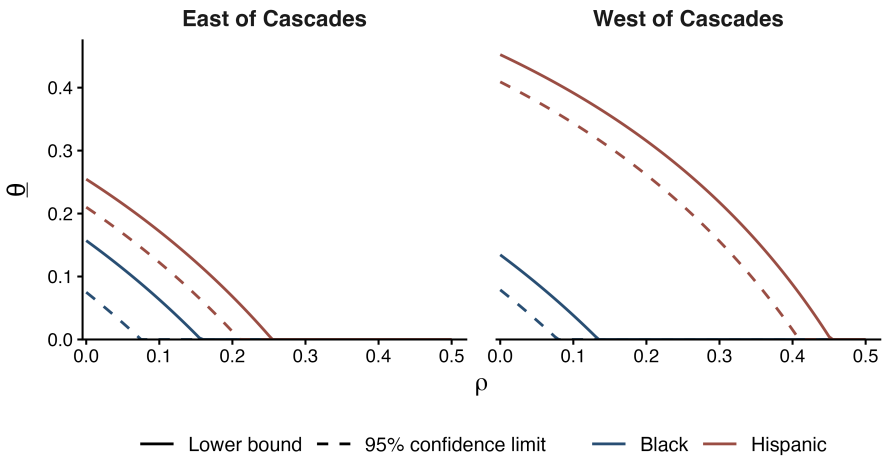


Figure 9: **Sensitivity to violations of Assumption 2, by side of the Cascade crest.** Solid lines are $\underline{\hat{\theta}}_{0,\rho}$ and dashed lines the one-sided 95% confidence limit.

D.2 Additional Tables

Table 7 reports the sensitivity of the lower bound when η and ρ are varied simultaneously. Figures 8 and 9 give the corresponding sensitivity curves for the geographic analysis discussed in Section 6.2, and Figures 10 and 11 the curves in κ for Native motorists.

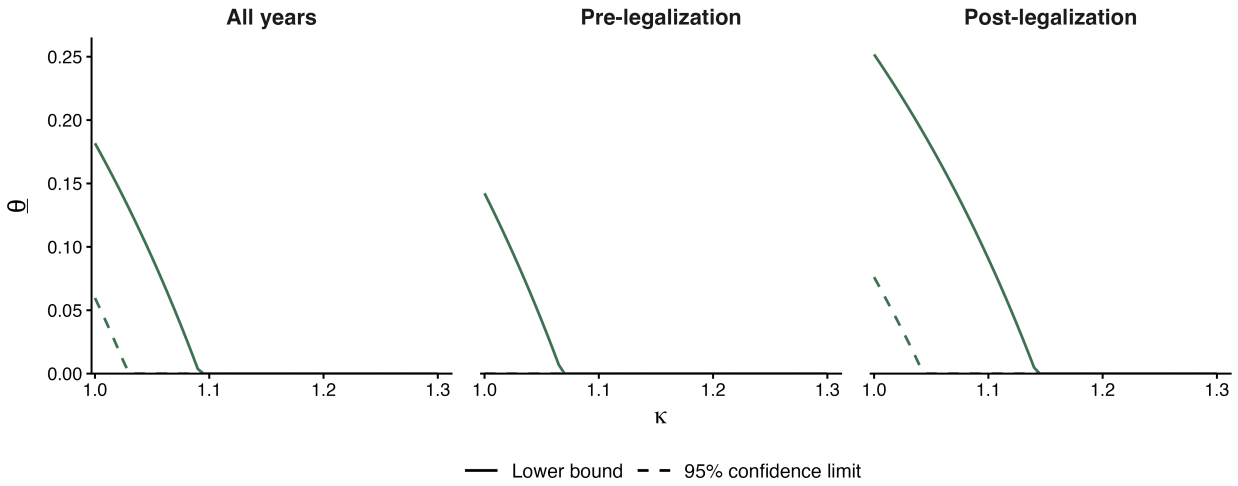


Figure 10: **Sensitivity of the lower bound to violations of Assumption 2, Native motorists.** Assumption 2 holds at $\kappa = 1$, so the horizontal axis measures departures from it in the direction consistent with Native hit rates exceeding White ones. Solid lines give $\underline{\hat{\theta}}_{0,\kappa}$ and dashed lines the one-sided 95% confidence limit, with η held at zero. The confidence limit before legalization is zero throughout and so coincides with the horizontal axis.

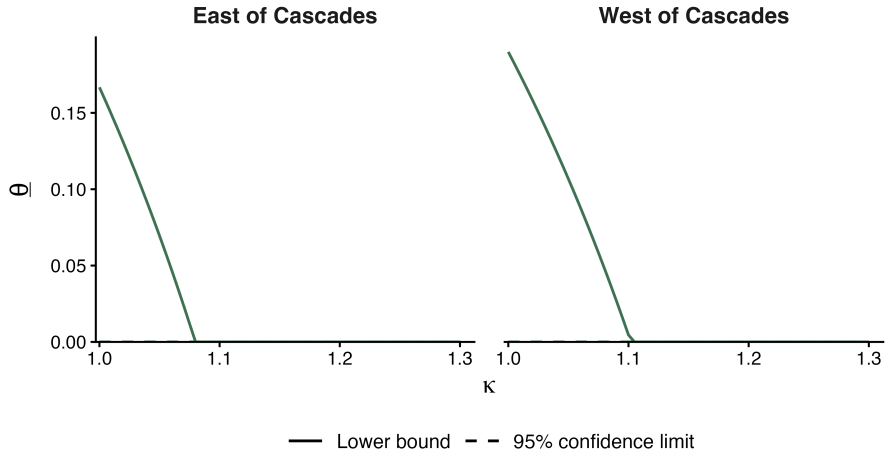


Figure 11: **Sensitivity of the lower bound to violations of Assumption 2 by region, Native motorists.** As in Figure 10, east and west of the Cascade crest. The confidence limit is zero throughout both panels and so coincides with the horizontal axis.

Table 7: Joint relaxation of both assumptions. Each entry gives the smallest violation of Assumptions 1 and 2 jointly at which the confidence limit on $\underline{\theta}$ reaches zero. For a given column η the number in the table is the corresponding value of ρ; an entry of 0.00 indicates that the limit has already reached zero under the violation of Assumption 1 alone. Native motorists are omitted because their bound rests on the miss branch, which ρ does not erode.

		η									
Race	Subgroup	0.00	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45
Black	All years	0.09	0.07	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Pre-legalization	0.06	0.04	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Post-legalization	0.10	0.07	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	East of Cascades	0.07	0.04	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	West of Cascades	0.08	0.06	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Hispanic	All years	0.29	0.27	0.24	0.21	0.18	0.14	0.08	0.01	0.00	0.00
	Pre-legalization	0.29	0.27	0.25	0.22	0.19	0.15	0.10	0.03	0.00	0.00
	Post-legalization	0.25	0.23	0.20	0.17	0.12	0.07	0.00	0.00	0.00	0.00
	East of Cascades	0.22	0.20	0.18	0.15	0.11	0.06	0.01	0.00	0.00	0.00
	West of Cascades	0.42	0.40	0.38	0.36	0.33	0.29	0.24	0.18	0.08	0.00